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Periodically Poled Ridge Waveguides in LiNbO_3

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To my family

Abstract

The present thesis concerns an optimized design for integrated nonlinear optical devices in Lithium Niobate ($LiNbO_3$) and includes their fabrication and characterization. Ridge waveguides grant a better confinement of light compared to channel waveguides and can therefore lead to more efficient and smaller integrated nonlinear optical devices. The main fabrication techniques used here are reactive ion etching (RIE), which is a dry etching process potentially scalable to the nanoscale, and annealed proton exchange (APE). After a separate optimization of these two critical processes, the full fabrication cycle of a device is closed by combining the APE RIE ridge fabrication process with periodically poled domain structures employed to bring to quasi phase matching (QPM) nonlinear optical interactions in the ridge waveguide.

The final optimized recipe for the RIE of $LiNbO_3$ developed in this work allows a significant improvement in etching slope angles of 83° to be compared with previous values around 60° and depths (up to $1\text{ }\mu\text{m}$ into the crystal). A thorough experimental study of the optical properties of annealed PE layers in undoped and MgO -doped $LiNbO_3$ substrates resulted in the development of an empirical model for the nonlinear proton diffusion during annealing, which then provided the basis for further optimizations towards the fabrication of a nonlinear optical demonstrator. The simulation outputs of the nonlinear diffusion modeling were used to analyze the transverse mode profiles and effective indexes in the waveguides and provided guidelines for the optimization of the fabrication process, targeting single-mode guidance of the fabricated ridges at telecom wavelengths. Finally, the properties of the fabricated periodically poled APE ridge waveguides devices were characterized by atomic force microscopy (AFM) and an automated setup was built for performing nonlinear experiments in them.

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Contents

Contents	i
1 Introduction	1
1.1 Aim	2
1.2 Outline	2
2 Fundamental Background and Methods	3
2.1 Waveguides and Nonlinear Optics	3
2.1.1 Nonlinear Optics Theory	3
2.1.2 Waveguides	10
2.1.3 SHG in a waveguide	11
2.2 Material Properties of Lithium Niobate	13
2.2.1 Dielectric Properties	14
2.2.2 Ferroelectric Properties	16
2.2.3 Photorefractive Effect	17
2.3 Proton Exchange	17
2.3.1 PE Process	19
2.4 Electric Field Poling	20
3 Device Design and Optimization	23
3.1 Improved Dry Etching of Lithium Niobate	24
3.1.1 Technology state of the art	24
3.1.2 Experimental Conditions	25

3.1.3	Results and Discussions	27
3.2	Study of Annealed Proton Exchanged Layers	31
3.2.1	Characterization	32
3.2.2	Modeling	33
3.3	Simulations	39
3.4	The Final Design	43
4	Device Fabrication and Characterization	45
4.1	Details of the Fabrication	45
4.2	Process Inspection	47
4.3	Optical Setup	50
4.4	Results and Discussion	53
5	Conclusions and Future Work	57
	Bibliography	59
A	Appendix Poling	65
	List of Figures	67
	List of Tables	69

Chapter 1

Introduction

Lithium Niobate ($LiNbO_3$) is a very important material in photonics and has been used to produce many different kind of devices - primarily because of its large electro-optic, nonlinear, photo-elastic, pyroelectric and piezoelectric coefficients [1, 2].

This project mainly aimed at the efficient exploitation of its nonlinear properties (which can produce effects like sum-frequency generation, optical parametric oscillation or frequency doubling) in integrated optical formats. There are many other crystalline materials which possess nonlinear optical properties, e.g. potassium titanyl phosphate, lithium tantalate etc., but also polymers [3] and III-V semiconductor materials like silicon [4], gallium nitride, gallium arsenide or indium phosphate.

Silicon is by far the most used material in the semiconductor area and on-chip photonics is rapidly becoming more and more important. Due to the fact that its nanotechnology platform is already very advanced, it is possible to fabricate very small waveguides and photonic crystals in it, which yield very high confinement. This is the reason why one can achieve high efficiencies of nonlinear effects even though its nonlinearity is much lower than in $LiNbO_3$. Thus, there is a need for a better technology platform for $LiNbO_3$ to keep it competitive for nonlinear optics applications. When higher confinement integrated photonic structures in this material are realized, the devices will be smaller, faster and more efficient. Applications for those kinds of devices are not only wavelength conversion (here) but also all-optical signal processing in telecom systems (phase-regeneration, chirp removal, retiming etc.) [5], optical switching [6] with devices

such as all-optical diodes [7] or transistors and many more.

1.1 Aim

The aim of this thesis was to develop high confinement waveguides in $LiNbO_3$. Therefore ridges were to be etched by reactive ion etching (RIE) into the crystal, which is particularly hard to etch. In order to achieve efficient guidance while preserving the high nonlinearity of the $LiNbO_3$ substrate, proton exchange (PE) combined with annealing was used, and hence the next task was to find optimized conditions for those processes to be then combined with optimized etching. This required an extensive experimental characterization and subsequent modeling of the annealing process. Moreover, the project was further expanded to also include fabrication of quasi phase matched (QPM) grating structures and their combination with ridge waveguides, with the aim of achieving efficient second-harmonic generation from wavelengths used in optical telecommunications, verifying interaction aspects of the fabrication process for such nonlinear integrated optical devices and performing first optical measurements on them.

1.2 Outline

The thesis is organized as follows. Chapter 2 will provide an overview of the theoretical background for the developed ridge waveguide devices (concerning nonlinear optics and optical waveguides) and will briefly summarize state-of-the-art fabrication methods in $LiNbO_3$. Chapter 3 will then explain the optimization of the reactive ion etching (RIE) process used to etch deep structures into $LiNbO_3$, the modeling of the annealing of PE layers and the simulations carried out to evaluate the guidance properties of the $LiNbO_3$ ridge waveguides. Finally, an optimized design will be given. This design was implemented by combining all the different technology steps and their optimization, as fully described in Chapter 4, which also includes results of the inspection of the final devices by AFM and optical techniques, and a presentation of the automated optical setup that was also constructed to perform nonlinear experiments with the fabricated devices. The thesis will end with a conclusion and further perspectives will be given.

Chapter 2

Fundamental Background and Methods

This chapter describes the basic theory and methods most relevant to this project. It summarizes the ideas behind the fabrication, simulations and experimental parts presented in the following two chapters.

2.1 Waveguides and Nonlinear Optics

Developing practical photonic devices requires guiding light in an appropriate way in a specific material for maximum compactness and efficiency. The following pages explain how the guiding can be achieved, what theoretically happens in optical nonlinear materials and which effects can result from this.

2.1.1 Nonlinear Optics Theory

This part explores the theory behind nonlinear optics and it will also explain briefly how to get from Maxwell's equations to the coupled wave equations describing second-harmonic generation (SHG) in an optical medium. The first nonlinear experiment was shown by Franken et al. [8] in 1961 and since then a lot of work has been done in this field. In order to be able to describe nonlinear optical phenomena one should first consider the dependence of the polarization on the electromagnetic field in matter. Once

an electric field $E(t)$ is applied to an optical medium, electric dipoles are formed in the material and this is macroscopically causing an induced polarization $P(t)$. Now there is a difference between a small and a strong electric field. In the case when $E(t)$ is rather small the motion of electrons from their original position can be considered to linearly follow the oscillations of the electric field. The induced polarization will be linear and can be described with:

$$P(t) = \varepsilon_0 \chi^{(1)} E(t) \quad (2.1)$$

where ε_0 is the dielectric constant in vacuum and $\chi^{(1)}$ is the first order susceptibility tensor, which is in general a complex quantity. Its real part determines the refractive index, while its imaginary part is related to losses. If the electrical field becomes larger, the linear dependence is no longer sufficient to describe the material response, because the electrons can not follow the oscillations of the applied field anymore. Their distorted motion generates new frequency components in the field. Such nonlinear effects can be accounted for by a Taylor series expansion of the polarization including higher order terms in equation 2.1:

$$P(t) = \varepsilon_0 \chi^{(1)} E(t) + \varepsilon_0 \chi^{(2)} E(t)E(t) + \varepsilon_0 \chi^{(3)} E(t)E(t)E(t) + \dots \quad (2.2)$$

where $\chi^{(2)}$ and $\chi^{(3)}$ are the second and the third order susceptibility. Only materials with a non-centrosymmetric crystal lattice can generate second order effects. In centrosymmetric crystals the even terms of equation 2.2 are equal to zero. The second order term $\varepsilon_0 \chi^{(2)} E(t)E(t)$ creates effects such as second-harmonic generation, sum-frequency generation, difference-frequency generation, optical rectification, optical parametric oscillation and the Pockels effect, while the third-order term $\varepsilon_0 \chi^{(3)} E(t)E(t)E(t)$ is responsible for third-harmonic generation, Brillouin scattering, Kerr effect and an intensity dependent refractive index.

The coupled wave equations

This paragraph concerns the derivation of the coupled wave equations, describing second order three wave mixing interactions. For the sake of simplicity the derivation is done for the bulk. Further details can be found in e.g. reference [9] or [10].

We are beginning from Maxwell's equations and considering a nonmagnetic, lossless, dispersionless medium with no free charges or free currents. We denote

$$\nabla \cdot D = 0 \quad (2.3)$$

$$\nabla \cdot B = 0 \quad (2.4)$$

$$\nabla \times E = -\frac{1}{c} \frac{\partial B}{\partial t} \quad (2.5)$$

$$\nabla \times B = \frac{1}{c} \frac{\partial D}{\partial t} \quad (2.6)$$

Calculating the curl of (2.5) leads to:

$$\nabla \times \nabla \times E = -\frac{1}{c} \frac{\partial}{\partial t} (\nabla \times B) \quad (2.7)$$

By using (2.7) with (2.6) and assuming $-\nabla^2 \cdot E \approx \nabla \times \nabla \times E$ we get:

$$\nabla^2 \cdot E = \frac{1}{c^2} \frac{\partial^2 D}{\partial t^2} \quad (2.8)$$

D is the displacement field which is given by:

$$D = \varepsilon_0 E + P \quad (2.9)$$

We denote the nonlinear part from equation 2.2 as P^{NL} :

$$P(t) = \varepsilon_0 \chi^{(1)} E(t) + P^{NL} \quad (2.10)$$

By inserting equation 2.10 into (2.9) we obtain:

$$D = \varepsilon_0 (1 + \chi^{(1)}) E + P^{NL} \quad (2.11)$$

Here $1 + \chi^{(1)}$ is the definition of the electric constant ε_r , which is related to the refractive index n of the medium by the following relationship:

$$n^2 = \Re[\varepsilon_r] = \Re[1 + \chi^{(1)}] \quad (2.12)$$

Substituting for (2.12) in equation 2.11 yields:

$$D = \varepsilon_0 n^2 E + P^{NL} \quad (2.13)$$

Using (2.13) in (2.8) leads now to:

$$\nabla^2 E(t) - \frac{n^2}{c^2} \frac{\partial^2 E(t)}{\partial t^2} = \frac{1}{c^2} \frac{\partial^2 P^{NL}}{\partial t^2} \quad (2.14)$$

Given that we are considering second-order nonlinear effects, $P^{NL} = \varepsilon_0 \chi^{(2)} E(t)^2$ in our case. We shall now move from the time domain to the frequency domain and consider monochromatic waves. Specifically, we shall write the total field as the sum of three components, at $\omega_1, \omega_2, \omega_3$:

$$E = \Re \{ E_1 e^{i\omega_1 t} + E_2 e^{i\omega_2 t} + E_3 e^{i\omega_3 t} \} \quad (2.15)$$

satisfying energy conservation:

$$\omega_3 = \omega_1 + \omega_2 \quad (2.16)$$

Depending on the input conditions this can correspond to sum-frequency generation (SFH), difference-frequency generation (DFG), etc. We are just having a look at an up-conversion process, namely sum-frequency generation (SFG) from the input waves at ω_1 and ω_2 :

$$\omega_1 + \omega_2 \rightarrow \omega_3$$

This process is sketched in figure 2.1.

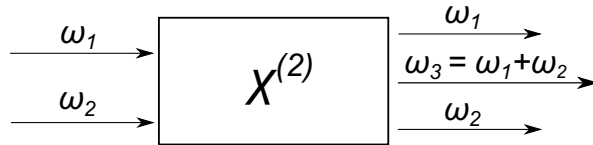


Figure 2.1: Sum-frequency generation

By converting $P^{NL} = \varepsilon_0 \chi^{(2)} E(t)^2$ to the frequency domain and isolating the polarization component at ω_3 , considering plane waves with $k_i = n(\omega_i) \frac{\omega_i}{c}$ while $i = 1, 2, 3$ and assuming that the envelope of the plane waves varies slowly with the propagation coordinate x (Slowly-Varying Envelope Approximation, short: *SVEA*), one gets from 2.14 to the

coupled wave equations [9]:

$$\frac{\partial E_1}{\partial x} = \frac{-i\omega_1}{n_1 c} d E_3 E_2^* e^{-i\Delta k x} \quad (2.17)$$

$$\frac{\partial E_2}{\partial x} = \frac{-i\omega_2}{n_2 c} d E_3 E_1^* e^{-i\Delta k x} \quad (2.18)$$

$$\frac{\partial E_3}{\partial x} = \frac{-i\omega_3}{n_3 c} d E_1 E_2 e^{i\Delta k x} \quad (2.19)$$

where $\Delta k = k_3 - k_2 - k_1$ is the *phase mismatch* and $d \propto \frac{\chi^{(2)}}{2}$ the nonlinear coefficient which depends on the polarization of the three waves and on the employed nonlinear crystal.

Second-harmonic generation (SHG)

SHG is now a special case of SFG with only one input frequency (the “fundamental”) $\omega_{FF} = \omega_1 = \omega_2$ and $E_{FF} = E_1 = E_2$. So, 2.16 reduces to:

$$\omega_{SH} = \omega_{FF} + \omega_{FF} = 2\omega_{FF} \quad (2.20)$$

where ω_{SH} is the frequency of the generated second harmonic. The corresponding geometry and the energy level diagram are depicted in figure 2.2.

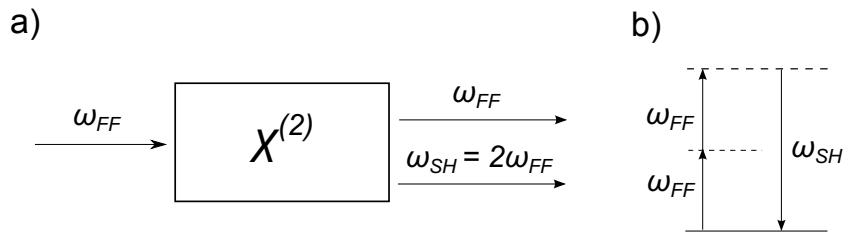


Figure 2.2: a) Geometry of second harmonic generation and b) energy level diagram

The coupled wave equations 2.17-2.19 then simplify into:

$$\frac{\partial E_{FF}}{\partial x} = \frac{i\omega_{FF}}{n_{FF} c} d E_{SH} E_{FF}^* e^{-i\Delta k x} \quad (2.21)$$

$$\frac{\partial E_{SH}}{\partial x} = \frac{i\omega_{FF}}{n_{SH} c} d E_{FF}^2 e^{i\Delta k x} \quad (2.22)$$

where the phase mismatch Δk is now:

$$\Delta k = k_{SH} - 2k_{FF} = \frac{2\omega_{FF}}{c} (n_{SH} - n_{FF}) \quad (2.23)$$

In the no pump-depletion approximation ($\frac{\partial E_{FF}}{\partial x} \sim 0$), by integrating directly equation 2.22 over the length L , the intensity of the second-harmonic wave I_{SH} can be obtained as:

$$I_{SH} = 2 \frac{\omega_{FF}^2 d^2 L^2}{\epsilon_0 n_{FF}^2 n_{SH} c^3} \left(\frac{\sin(\Delta k L / 2)}{\Delta k L / 2} \right)^2 I_{FF}^2 \quad (2.24)$$

where I_{FF} is the intensity of the pump at the fundamental frequency and L the interaction length. The refractive indexes n_{FF} and n_{SH} are in general different because of the dispersion in the material.

The quadratic factor with the sinus in equation 2.24 suggests that the highest intensity is achieved at $\Delta k = 0$ a condition corresponding to a perfect phase matching. Usually, SHG is characterized by the coherence length L_c which is defined as follows:

$$L_c = \frac{\pi}{\Delta k} \quad (2.25)$$

and represents the maximum length within which the generated second-harmonic field amplitude is growing monotonically. A high mismatch means small coherence length, then E_{SH}^2 oscillates in propagation without yielding efficient SHG at the output (figure 2.4, b). On the other hand, perfect phase matching corresponds to $L_c \rightarrow \infty$ (figure 2.4, a).

Phase matching

Phase matching is a crucial point in achieving highly efficient parametric processes. From equation 2.24 it is clear that $\Delta k = 0$ would require $n_{FF} = n_{SH}$, which, due to dispersion in the material, is generally not possible. However, several techniques have been developed to overcome this problem. The first is called *birefringent phase matching* (BPM) because it uses the birefringence in anisotropic crystals to bring Δk to zero (equation 2.23) by using different polarizations for the fundamental frequency and its second-harmonic. Nevertheless, BPM is restricted to specific wavelength combinations only, which is not particularly practical. Besides, not all crystals possess sufficient birefringence to compensate dispersion. Another possibility is to engineer the nonlinear coefficients of the crystal in a periodic manner, which is known as *quasi phase matching* (QPM). The idea behind it is to correct the phase mismatch by a spatial modulation of the nonlinear susceptibility $\chi^{(2)}$. In ferroelectric crystals as $LiNbO_3$ this can be realized

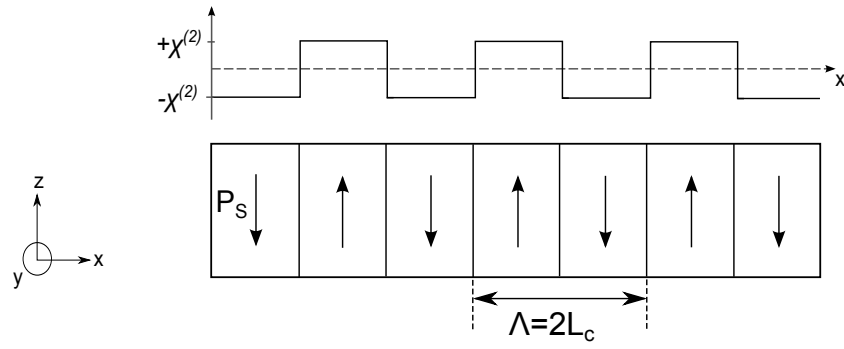


Figure 2.3: Pattern of periodically engineered spontaneous polarization in a ferroelectric crystal for first order QPM. The arrows marking the orientation of the spontaneous polarization.

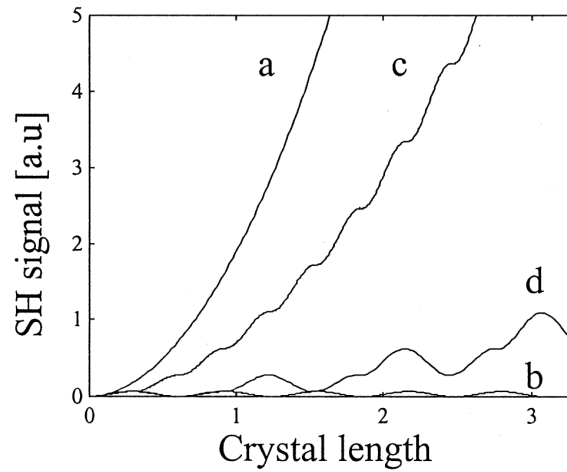


Figure 2.4: Second-harmonic power growth depending on the distance the fundamental wave propagated through a nonlinear medium for a) perfect phasematching, b) no phasematching c) first order QPM and d) third order QPM. [11]

by reversing periodically the sign of the spontaneous polarization. In case of SHG the fundamental and second-harmonic waves possess different phase velocities and gain a phase mismatch of π over the coherence length L_c . If the spontaneous polarization is engineered periodically along the propagation direction like in figure 2.3 a phase shift of π after each half period ($\Lambda/2$) is artificially added to the SHG interaction, meaning that the phase mismatch becomes zero again. So, for first order QPM the period of the ferroelectric domain grating has to be:

$$\Lambda = 2L_c \quad (2.26)$$

Higher-order QPM is also possible, but less efficient. For instance third order QPM can be realized by switching the domain every three coherence lengths. The resulting power of the second-harmonic is depicted in 2.4, c and d for first and third order, respectively. With QPM, SHG is modeled by the same coupled wave equations (2.22), but with an “effective” mismatch Δk_{QPM} :

$$\Delta k_{QPM} = \frac{2\omega_{FF}}{c}(n_{SH} - n_{FF}) - \frac{2\pi}{\Lambda}m \quad (2.27)$$

where m is the order of QPM. The nonlinear coefficient d appearing in the coupling constant of equations 2.21-2.22 is also replaced by an “effective” nonlinear coefficient d_m , which depends on the type of modulation.

2.1.2 Waveguides

The main tasks of this thesis is to develop technological, experimental and design tools to achieve high confinement optical waveguides in lithium niobate in connection with nonlinear applications. To make a waveguide one can use a layer structure of materials with different refractive indexes consisting of a guiding (n_G), a substrate (n_S) and a cladding (n_C) layer. The light is always guided in the layer with the highest refractive index (n) and that is why $n_C < n_S < n_G$ must be fulfilled. To be able to guide light at a given wavelength λ , the guiding layer needs to have a certain thickness. Due to Snell’s law, light will be reflected at the border to a lower index material and when incident at an angle larger than the critical angle, it will be trapped inside the layer because of total internal reflection. Depending on their geometry, one can have different types of waveguides. 2-D waveguides are only confining light in one transverse dimension, while 3-D waveguides (also called channel waveguides) are confining light in two transverse dimensions. The latter can be fabricated as buried waveguides or ridges as shown in figure 2.5. The ridge is expected to provide a stronger confinement, just because the low-index cladding is wrapping a larger portion of the waveguide.

There are different technologies to fabricate a higher index layer into a lithium niobate crystal, such as proton exchange (PE) [12], titanium diffusion [13], ion implantation [14] or direct ultraviolet writing [15]. I used PE together with etched ridges to achieve higher confinement for my waveguides. The PE method will be described later in section 2.3.

In the simplest case, waveguides have a step index profile. However, most of the time when the index change is achieved by a modification of the substrate, the waveguides have a graded index, where the index changes gradually in depth (and/or width). [16, 10]

Waveguide Theory

A 3D waveguide, defined by a given refractive index distribution $n(x, y)$, can sustain a discrete and finite number of guided modes, found as eigensolutions of the wave equation (2.14). The mode profile and the effective index can be found by numerically solving the following eigenfunction:

$$\left[\nabla^2 + \mu_0 \varepsilon_0 \omega^2 n^2(x, y) \right] F(x, y) = \beta^2 F(x, y) \quad (2.28)$$

where $n(x, y)$ is the refractive index profile and $F(x, y)$ is the transverse mode profile. We shall denote the wavefunction (field distribution) of the mode of order (m, n) as:

$$E^{(m,n)} = q_{m,n} F^{(m,n)}(x, y) e^{-i\beta^{(m,n)}z} \quad (2.29)$$

where $q_{m,n}$ is a normalization term and $\beta^{(m,n)}$ is the propagation constant, defined by:

$$\beta^{(m,n)} = \frac{2\pi}{\lambda} N_{eff}^{(m,n)} \quad (2.30)$$

where $N_{eff}^{(m,n)}$ is the effective mode index.

2.1.3 SHG in a waveguide

For guided-wave SHG some of the conditions for the coupled wave equations are changing and they have to be derived again. Without providing the full details of the derivations, this paragraph just summarizes the most important differences in the resulting

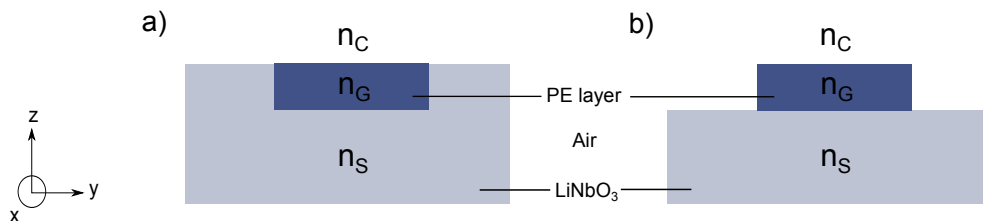


Figure 2.5: a) Buried and b) ridge waveguide structure

equations. I have chosen to describe this only for the interaction between one pair of modes - namely TM modes of order $m=0$ and $n=0$, in which case $N_{FF} = N_{eff}^{TM_{00}}$ at the fundamental frequency and $N_{SH} = N_{eff}^{TM_{00}}$ at the second-harmonic.

The nonlinear coupling coefficients in the equations 2.21-2.22 are getting more complex to account for the spatial overlap of the interacting modes. In this case equation 2.22 becomes to:

$$\frac{\partial E_{SH}}{\partial x} = i\omega_{FF}\kappa E_{FF}^2 e^{i\Delta\beta z} \quad (2.31)$$

where the *nonlinear coupling coefficient* κ is given by:

$$\kappa = d \sqrt{\frac{2\mu_0}{cN_{FF}^2 N_{SH}}} \sqrt{\frac{1}{S_{ovl}}} \quad (2.32)$$

where S_{ovl} is the *mode overlap*, which is defined as:

$$S_{ovl} = \frac{\left(\int |F_{FF}|^2 dx dy\right)^2 \int |F_{SH}|^2 dx dy}{\left(\int F_{FF}^2 F_{SH} dx dy\right)^2} \quad (2.33)$$

By replacing k with β for the given mode pair, the phase mismatch (equation 2.23) becomes:

$$\Delta\beta = \beta_{SH} - 2\beta_{FF} \quad (2.34)$$

To compensate for this phase mismatch by first order QPM, one has to introduce a domain grating with the period Λ that fulfills equation 2.26. As a result one gets the condition:

$$\Delta\beta - \frac{2\pi}{\Lambda} = 0 \quad (2.35)$$

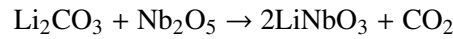
By putting (2.30) into (2.34), then into (2.35) and changing the equation, the grating period becomes finally:

$$\Lambda = \frac{\lambda_{FF}}{2|N_{SH} - N_{FF}|} \quad (2.36)$$

where λ_{FF} is the wavelength of the fundamental wave, N_{SH} and N_{FF} are the effective indexes of the second-harmonic and the fundamental wave, respectively for the involved modes.

2.2 Material Properties of Lithium Niobate

Lithium niobate is one of the most important materials for integrated optics and is widely used for many different devices such as second-harmonic generators, acoustic wave transducers, beam deflectors, optical modulators, phase conjugators and many more. It is a non-centrosymmetric, negative uniaxial and ferroelectric crystal, which is also characterized by large pyroelectric, piezoelectric, nonlinear, photo-elastic and electro-optic coefficients.[1, 2] It does not exist in nature and is instead grown by using the Czochralski technique from a melt of lithium carbonate (Li_2CO_3) and niobium pentoxide (Nb_2O_5). The reaction is described as follows:



Only *stoichiometric lithium niobate* (SLN) is matching exactly the chemical compound symbol, LiNbO_3 , with a ratio of [Li]:[Nb]:[O] of 1:1:3. *Congruent lithium niobate* (CLN) is more commonly used instead, because it can be more easily grown. Here the total ratio of *Li* to the total *Li* and *Nb* concentration is 48.45% . [17]

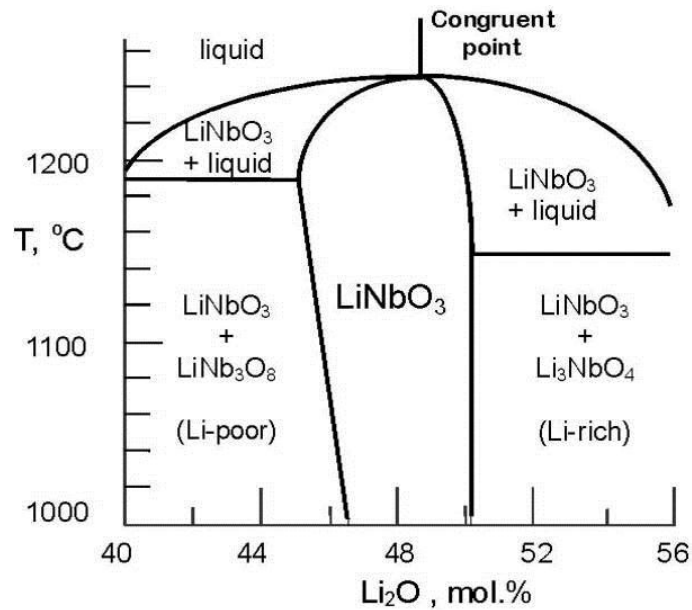


Figure 2.6: Composition phase diagram of LiO_2 - NbO_5 compound [17]

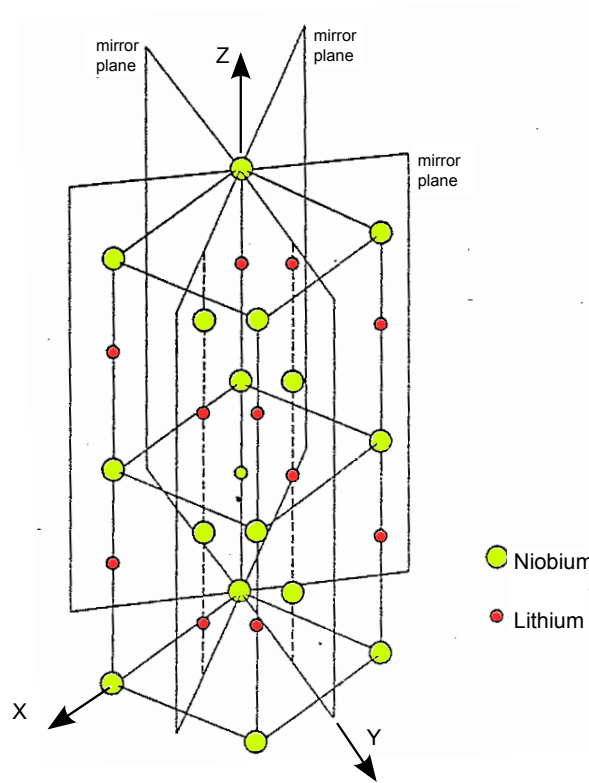


Figure 2.7: Standard orientation of X,Y,Z axes, after Weis and Gaylord [1]

Lithium Niobate is an uniaxial crystal and belongs to the *perovskite* family. More detailed information on the crystal structure can be found in Weis and Gaylord [1]. Most important here is the convention used for its crystallographic axes, which is shown in figure 2.7.

The following paragraphs will explain some of the main physical properties of LiNbO_3 , which are needed to understand basic techniques and approaches involved in my project.

2.2.1 Dielectric Properties

Lithium niobate is a dielectric, anisotropic material and the relation between the electric field E and the electric displacement D is accordingly given by:

$$D_i = \sum_j \epsilon_{ij} E_j \quad i, j = x, y, z \quad (2.37)$$

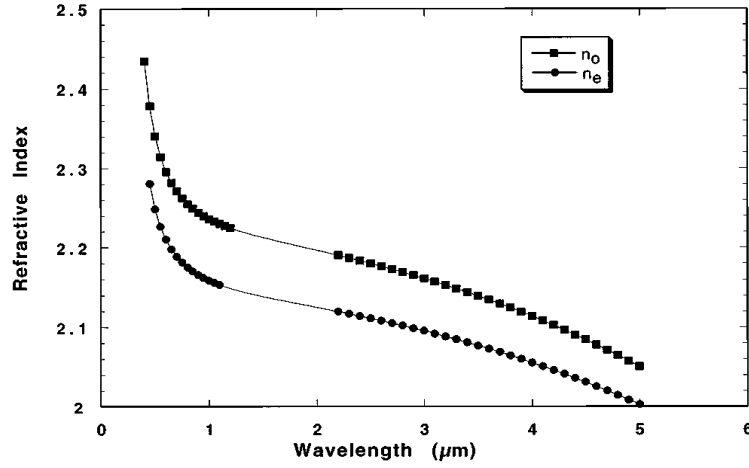


Figure 2.8: Dispersion curves for the ordinary and extraordinary index of CLN [18]

where ε_{ij} is the ij^{th} component of the dielectric tensor with:

$$\varepsilon_{ij} = \varepsilon_0(\varepsilon_r)_{ij} \quad (2.38)$$

where ε_0 is the dielectric constant of vacuum and $(\varepsilon_r)_{ij}$ is the ij^{th} component of the relative permittivity. In the reference frame (principal axes) of figure 2.7 the dielectric tensor can be displayed in a matrix of diagonal form:

$$\varepsilon_{ij} = \begin{pmatrix} \varepsilon_{11} & 0 & 0 \\ 0 & \varepsilon_{22} & 0 \\ 0 & 0 & \varepsilon_{33} \end{pmatrix} \quad (2.39)$$

where $\varepsilon_{11} = \varepsilon_{22} \neq \varepsilon_{33}$, so $LiNbO_3$ is an *anisotropic, uniaxial* material. Taking equation 2.12 into account we can also relate it to the refractive indexes:

$$n_o = n_x = n_y = \sqrt{\frac{\varepsilon_{11}}{\varepsilon_0}} \quad n_e = n_z = \sqrt{\frac{\varepsilon_{33}}{\varepsilon_0}} \quad (2.40)$$

where n_o is the index along the X and Y axes denoted as the *ordinary* refractive index and n_e is the *extraordinary* index (along the Z-axis). Since $n_e < n_o$ the crystal is *negative uniaxial*.

The dispersion in $LiNbO_3$ can be described by Sellmeier equations. The following formula is valid for undoped CLN at 21°C [18]:

$$n^2 = 1 + \frac{A\lambda^2}{\lambda^2 - B} + \frac{C\lambda^2}{\lambda^2 - D} + \frac{E\lambda^2}{\lambda^2 - F} \quad (2.41)$$

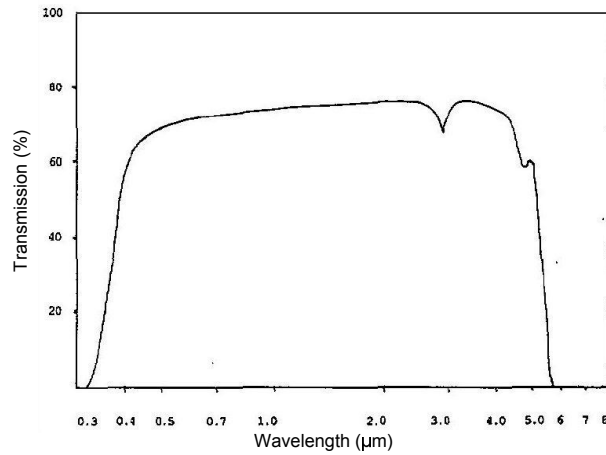


Figure 2.9: Transmission spectrum of CLN [19]

where λ is the wavelength. Using the *Sellmeier coefficients* ($A-F$) given by Jundt/Zelmon [18], we can obtain the dispersion curves within the material transparency window shown in figure 2.8. The transparency window is shown additionally in figure 2.9. As can be seen is the transparency range of LiNbO_3 quite large.

2.2.2 Ferroelectric Properties

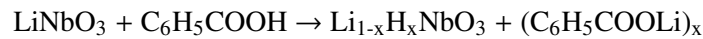
At low temperatures LiNbO_3 is ferroelectric, but once over its Curie temperature ($T_C = 1143^\circ\text{C}$) it changes to a paraelectric phase. At room temperature the crystal possesses permanent dipole moments causing a spontaneous polarization P_S . A domain is a region where such dipoles are uniformly oriented (constant P_S). It can spontaneously arise in a crystal during growth, temperature or chemical treatment. Often ferroelectric domains are randomly distributed and represent an unwanted effect as they reduce the average piezoelectric, electro-optical and nonlinear optical coefficients. Nevertheless, domains can also be engineered by a technique called *field electric poling* (EFP), which will be described later. The polarization is determined by the position of Li^+ and Nb^+ ions with respect to the oxygen layers. [17] Single-domain wafers can be fabricated by applying a voltage during the Czochralski process [2].

2.2.3 Photorefractive Effect

The photorefractive effect consists in optically induced changes of the refractive index and is mainly caused by the bulk photovoltaic effect [1]. Light passing through $LiNbO_3$ can generate free electrons, mostly coming from impurities, mainly iron ions. These photo-excited electrons tend to accumulate locally and create via the eletro-optic effect local changes of the refractive index. This effect is generally not wanted and can be overcome by (a) operating at higher temperatures to increase the photoconductivity, (b) making the crystal more stoichiometric, in order to reduce the presence of impurities, or (c) doping $LiNbO_3$ with e.g. magnesium monoxide (MgO), which acts as an acceptor for the free electrons. [17]

2.3 Proton Exchange

In order to fabricate waveguides into $LiNbO_3$ one needs a technique to increase locally its refractive index. One of the most common techniques used for this purpose is *proton exchange* (PE), which leads to chemical modifications of the crystal structure, basically consisting in the substitution of Li^+ ions with protons (H^+ ions) at the surface. This is generally done by immersing the crystal in an acid at temperatures around 150-300°C. The formation of waveguides by PE was first reported by Jackel et al. in 1982 [12]. PE increases the extraordinary index only, while the ordinary index is decreased. Thus, it makes it possible to confine TM-modes in Z-cut and TE-modes in X or Y-cut $LiNbO_3$. [12] There are many acids which can be used for PE, but the most used one is benzoic acid (BA, C_6H_5COOH), which is a rather weak acid (therefore not dangerous and easier to handle) with a low melting point around 150°C. The reaction giving rise to proton-rich layers can be described as follows:



where x is the substitution rate of hydrogen, which is about $x \approx 0.7$ for our conditions. The waveguides obtained by direct PE in BA have typically a step index profile. The increase of the extraordinary refractive index is about $\Delta n_e = 0.12$ at $\lambda = 633 \text{ nm}$ and the decrease of the ordinary index $\Delta n_o = -0.04$. [20] In MgO-doped $LiNbO_3$ the value

for the extraordinary index increase is only slightly higher [21]. PE is based on a diffusion process and accordingly the resulting thickness of the exchanged layer d_{PE} can be obtained as [10]:

$$d_{PE} = 2\sqrt{D_{PE}(T_{PE})t_{PE}} \quad (2.42)$$

where T_{PE} is the temperature, t_{PE} is duration of PE and D_{PE} is the proton diffusion coefficient. Equation 2.42 means that D_{PE} is temperature-dependent. Its dependence is given by an Arrhenius law:

$$D_{PE} = D_0 \exp\left\{-\frac{E_a}{kT_{PE}}\right\} \quad (2.43)$$

where D_0 is the diffusion constant, E_a the activation energy and k the Boltzmann constant ($k = 8.617343 \cdot 10^{-5} \text{ eV K}^{-1}$). An example of such a step index is shown in figure 2.10.

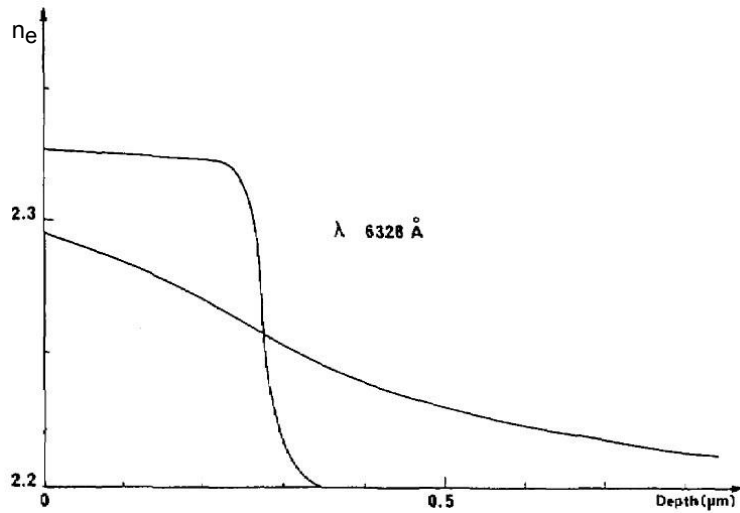


Figure 2.10: Refractive profiles for PE (step profile) and APE waveguides[22]

$Li_{1-x}H_xNbO_3$ possesses a very complex chemical structure and there has been a higher number of phases identified. The phase transitions can be characterized by abrupt variations of the refractive index (Δn) and/or the crystal deformation (ϵ_{33}'' , stress tensor). The different phases are indicated with $\alpha, \kappa_1, \kappa_2, \beta_1, \beta_2, \beta_3, \beta_4$ and depicted in figure 2.11. The α -phase is the one closest to $LiNbO_3$ and supposed to generate the lowest losses and lead to almost no nonlinearity degradation. [23] So, for good nonlinear waveguides one has to ideally get into the α -phase. To recover the degraded nonlinear properties induced by

the PE, the waveguides have to be subsequently annealed, yielding graded index profiles such as the one shown in figure 2.10.

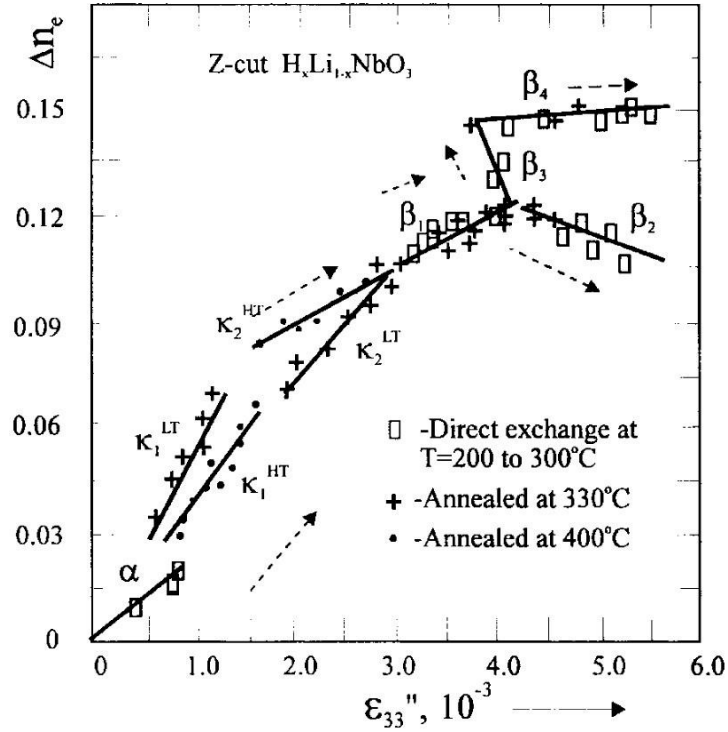


Figure 2.11: Structural phase diagram of $Li_{1-x}H_xNbO_3$ on Z-cut[23]

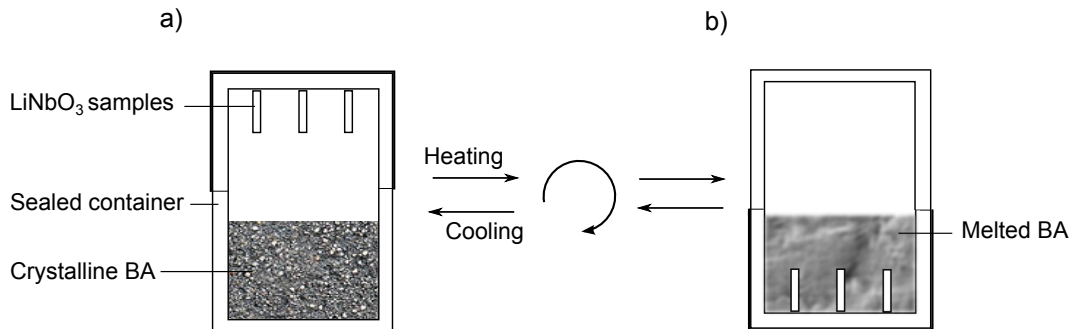


Figure 2.12: Schematic of the used PE setup

2.3.1 PE Process

The process of PE is relatively easy. Samples of $LiNbO_3$ are mounted on the top of a container and the bottom half is filled with BA powder (figure 2.12a). The setup is then

put into an oven and heated up to a given temperature above the BA melting point. Once the desired temperature is reached and the BA is melted completely, the setup is turned upside down and the $LiNbO_3$ samples are immersed in the acid (figure 2.12b). In order to stop the PE, the whole setup is turned again and then cooled down.

2.4 Electric Field Poling

This technique is based on the ferroelectric properties of $LiNbO_3$ described in section 2.2.2 and allows local switching of the spontaneous polarization P_S by applying an external electric field E_{ext} , which should exceed a threshold value called *coercive field* E_c . It has been first shown by Yamada et al. in 1993 [24]. EFP can be used to create the domain gratings needed for QPM and it is relatively easily to produce by this means Periodically Poled $LiNbO_3$ (PPLN) [25] with periods $\Lambda \geq 10 \mu m$. Submicron poling would also be possible, but the domains would be not very deep. Hence, there is an interest in high-confinement waveguides, which could confine light and exploit such tiny domains. In order to reverse a domain of an area A , the required amount of charge to be transferred to the crystal (Q_P) is given by:

$$Q_P = 2AP_S \quad (2.44)$$

Since conduction in CLN is negligible, the charge can also be expressed as:

$$Q_P = \int I_{pol} dt \quad (2.45)$$

where I_{pol} is the electric current measured during the poling, which provides a monitoring tool for the poling process. [26] The external poling field has to be applied along the Z-axis, because the polarization is oriented in this direction. For CLN $P_S = 0.78 \mu C/cm^2$ and $E_c = 21 kV/mm^2$.

Normal photolithography can be used to define the periodic electrode pattern. In this work I have used a patterned photoresist (without any metal electrodes), which acts as an insulator. The electrical contacts to the external circuit, through which E_c is applied, are made through a gel electrolyte in the openings of the photoresist. For 1D gratings the electrode pattern is adjusted to be parallel to the Y-axis, because the domains are grow-

ing much faster in this direction. A schematic result of a poled crystal with stabilized domains is shown in figure 2.13.

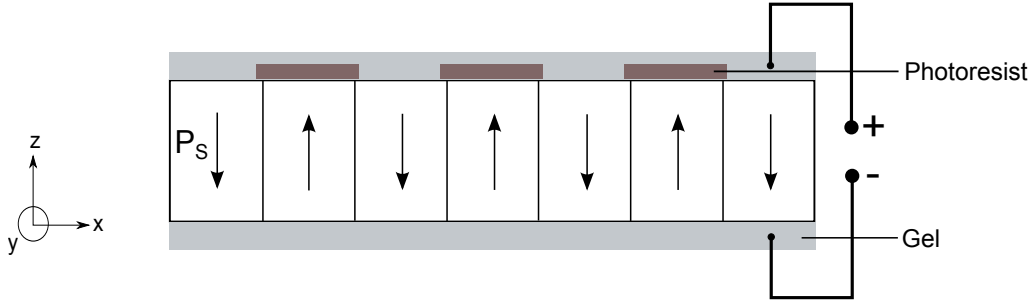


Figure 2.13: Schematic of PPLN with stabilized domains, on top: resist pattern used as mask and gel electrolyte for achieving electric contact to the crystal

After the poling the domains are usually only visible through crossed polarizers, but once the surface is etched one can see them under a microscope, because the different polarities ($\pm Z$) exhibit different etch rates [27] and therefore a surface relief pattern is produced. All in all, EFP can be challenging and it is still a current research topic, especially for short periods and 2D patterns. However, the EFP itself was not a main part of this particular project, which used relatively long periods in CLN and for which I used the standard in-house process of the group at KTH with the assistance of M. Manzo. A detailed description of the used EFP setup can be found in the thesis by M. Manzo, 2010 [26], but it consists mainly of a pulse generator, a high voltage amplifier, a simple electric circuit, an oscilloscope and a computer for tracking the signals. More details about the process and the domain growth mechanisms can be found in Miller, 1998 [28].

Chapter 3

Device Design and Optimization

The sections below contain a series of steps, which were first to be addressed in order to be able to fabricate the ultimate device presented in chapter 4. I fabricated ridge waveguides with periodical poling (PP) and the optimization had to start with the realization of ridge structures into $LiNbO_3$, but first the etching had to be improved. As a matter of fact, $LiNbO_3$ is an inert and relatively hard material and its structuring is still a big challenge. That is why a short introduction is provided on the current state of the technology in paragraph 3.1.1. The starting point for the etching recipes developed for this project were recently published results by other groups, which I have further optimized. After a good recipe was found, I fabricated a first set of waveguides. They possessed a guiding layer made by proton exchange (PE) with a thickness of 700 nm that was annealed afterwards. Unfortunately, they occurred to be multi-mode waveguides at the target operation wavelength ($\lambda = 1550$ nm). Single-mode-guidance is an essential feature to maximize the efficiency of our applications (SHG), so I set out to identify suitable conditions for that in a second waveguide optimization trial. In order to be capable of making simulations of the guidance properties, one needs full understanding of the behavior of the PE and of the subsequent annealing process. The aim is to be able to predict the index profile of the final waveguide for given PE and annealing conditions and hence provide guidelines for the next waveguide fabrication runs. The full characterization and modeling of the annealing will be given in chapter (3.2).

Once a suitable model is found, one estimates the profiles of the refractive index expected for given PE and APE conditions and then feed them into additional simulation software to check the guidance properties for different conditions and dimensions of the device. These modeling steps significantly speed up the ridge waveguide optimization process and help in identifying the most suitable fabrication conditions to achieve the desired performance. Furthermore the linear simulation tools make it possible to draw conclusions on the conditions required to quasi-phase-match SHG in the waveguide, hence helping in the choice of suitable grating periods for the poling step. The full design of such a device combining PPLN and ridge waveguides will be presented at the end of this chapter. The fabrication of the final device will be explained in chapter 4.

3.1 Improved Dry Etching of Lithium Niobate

Etching $LiNbO_3$ requires a fluorinated gas or acid. For our purpose the etching process is essentially characterized by two parameters. The first is the etching rate r , which is defined by:

$$r = \frac{d_{et}}{t_{et}} \quad (3.1)$$

where d_{et} is the measured etched depth and t_{et} the etching time. The second parameter is the selectivity s :

$$s = \frac{r_S}{r_M} \quad (3.2)$$

where r_S is the etching rate of the substrate and r_M the etching rate of the mask used to protect the areas which we do not want to be etched. Ideally we want a high selectivity, but this means the masking material should withstand the etching recipe used as much as possible, while maximizing the $LiNbO_3$ etching rate.

3.1.1 Technology state of the art

$LiNbO_3$ is a material of high interest for application to photonic devices and acoustic filters and there is consequently also a high interest in creating all kind of structures into this crystal. Hence, there is already a large number of publications on the structuring, yet challenges still remain in particular in as much as etching high aspect-ratio structures

for photonic applications is concerned. Initially, the concentration was mainly on wet etching with a mixture of hydrofluoric and nitric acid [29]. Nevertheless, wet etching instead is not a good solution for waveguides, because it is very isotropic and the sidewalls of a waveguide should be straight and as deep as possible. Some groups developed a method to cut deep ridges into $LiNbO_3$ by using a diamond saw [30, 31]. This is surely a highly engineered method and proved to be appropriate for fabricating waveguides in the micrometer regime. However, nanostructuring is definitely not possible with this method. Like in normal semiconductor technology $LiNbO_3$ can also be dry etched and this promises to make smaller structuring possible. Dry etching technologies that have been used so far are, among others, reactive ion etching (RIE) [32, 33, 34], ion beam milling [35], focused ion beam milling [36] and ion beam enhanced etching, which requires a step of wet etching, however [37].

The easiest to handle technique would be RIE and many different gases have been already used with this technique, the favorites being mixtures of SF_6 or CF_4 and Ar or He [33, 34]. The best results so far have been published by Jun et al. [34] and were obtained with inductive coupled plasma (ICP) and a gas mixture of SF_6 and Ar . They were achieving deep etching and also rectangular etched structures with a slope angle of 90° . Some of their main conclusions are that the chamber pressure should be very low, no cooling should be used and there should be no break in the etching process. In common with others they also used selective PE in order to increase the etching rate, because $Li_{1-x}H_xNbO_3$ was shown to be etched much faster [32].

3.1.2 Experimental Conditions

Apart from the preparation of the $LiNbO_3$ substrate materials, this whole process consists of three main parts: the fabrication of a chromium (Cr) hard mask (including chromium evaporation, photolithography and chromium dry etch), the actual optimization of a recipe for the etching of the substrate and the validation of the whole process. The etching studies were performed here on samples of pure CLN, PE CLN, APE CLN and z-surfaces of opposite polarity. How all of the above affect the results of the etching will be shown later.

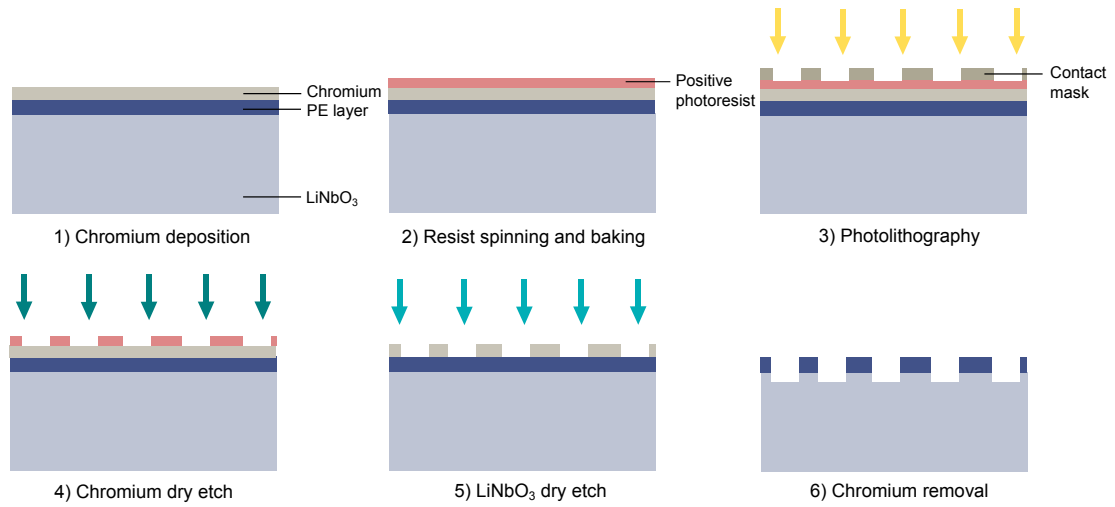


Figure 3.1: Fabrication of ridges into LiNbO_3

Figure 3.1 shows a sketch of the whole process for PE CLN samples. The thickness of the PE layer was 700 nm for this optimization. About 120 nm of chromium (Cr) were evaporated on the samples with an Edwards FL400 E-gun depositing system, with an evaporation rate of 0.1 nm/sec. Afterwards the positive photoresist S1818 was spin-coated and baked. Then photolithography was applied with a Karl Süss mask aligner model MJB3. This produced ridges of photoresist with a height of 1.5 μm on the Cr.

The dry etching was done with an Oxford Plasmalab 100 system, which is an ICP-RIE with helium backside cooling. For the Cr etch a mixture of Cl_2 and O_2 was used. Cooling was not needed during this process and this recipe had been already optimized within the group before. The time was chosen to slightly overetch the Cr for about 1 min. After this step the photoresist was removed with acetone and the ridges were measured with a surface profiler (KLA Tencor P-15). The used recipe turned out to be very selective. All in all this process step did not require much optimization and was quite stable.

The next step was the most challenging one and the one I had to fully optimize: the dry etching of the crystal. Based upon former publications by other groups I first performed tests with different gases (CF_4 , SF_6 and CHF_3). The same ratio of etching gas and argon was always used as for the RF and ICP powers. I additionally studied the influence of the chamber pressure and the cooling after finding out that in our case cooling was necessary in order to get good results.

In the first run I etched the samples only for one minute, while later I applied deep etching for about 20 min. After etching I measured the ridges with the profiler. I removed the Cr and measured again the ridge in order to obtain the etching rate of the Cr and the crystal. Additionally, I etched PE MgO:CLN with exactly the same recipe in order to be able to make comparisons with standard LN. (MgO:CLN is particularly interesting for high-confinement waveguides because it exhibits a reduced photorefractive effect.)

3.1.3 Results and Discussions

After several initial etching tests with different gases I could decide quickly on which gas to use for further optimizations. The results of those first runs are summarized in table 3.1. As it can be seen, CF_4 exhibits a high etching rate, but it is also etching the chromium very quickly and has therefore a lower selectivity (CF_4 table 3.2). SF_6 has instead a high selectivity and is therefore the gas I finally chose. CHF_3 is somewhere in the middle of both other gases. From the given tables one can also observe that the etching rates for PE CLN are much higher. The Li^+ ions are replaced with H^+ in such substrates, which prevents the redeposition of LiF . The latter is assumed to be the main reason why it is so hard to etch LiNbO_3 .

Crystal	SF_6 / Ar	CF_4 / Ar	CHF_3 / Ar
LN +Z	36	63.5	36.6
LN -Z	44.1	66.9	42
PE LN +Z	137.8	220.1	148.2
PE LN -Z	143.5	206.7	134.7

Table 3.1: Etching etching rate [nm/sec] with different gases

Crystal	SF_6 / Ar	CF_4 / Ar	CHF_3 / Ar
LN +Z	3.3	1.76	2.01
LN -Z	4.05	1.85	2.31
PE LN +Z	12.64	6.1	8.14
PE LN -Z	13.17	5.73	7.36

Table 3.2: Etching selectivity with different gases

In my first deep etching attempts with a longer time, I ran into problems with redeposition of chromium. In their study, Jun et al. [34] concluded that using no cooling or

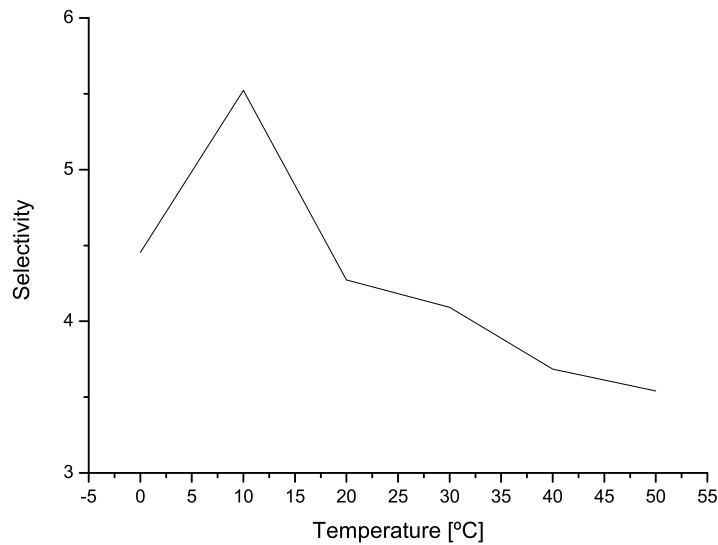


Figure 3.2: Influence of the temperature on the etching of -Z CLN

breaks during the etching would lead to the highest etching rates, yet this did not apply to my process. There was always a redeposition of chromium on the sample when long RIE runs were performed. Only cooling was able to solve this problem. Consequently, I had to optimize the cooling as well and studied the impact of different temperatures on the etching of -Z CLN. The result of this study is depicted in figure 3.2. The diagram shows that the highest selectivity is achieved at 10°C. I performed this study for deep etching, meaning that I etched the samples until there was almost no masking material (Cr) left.

In the end the deepest etched ridges had a height of 1 μm . An example of a measurement with the profiler is given in figure 3.3. The trenching which can be observed on both sides of the ridges is caused by ions glancing off the sidewalls of the ridge, resulting in an enhanced etching at the edges [34]. Figure 3.4 shows a microscope picture of the fabricated ridges.

In any case, the etching was a very sensitive process. During several attempts I recognized that keeping the chamber pressure very low is of paramount importance for achieving high etching rates. An slight increase in the pressure by just 1 or 2 mTorr is

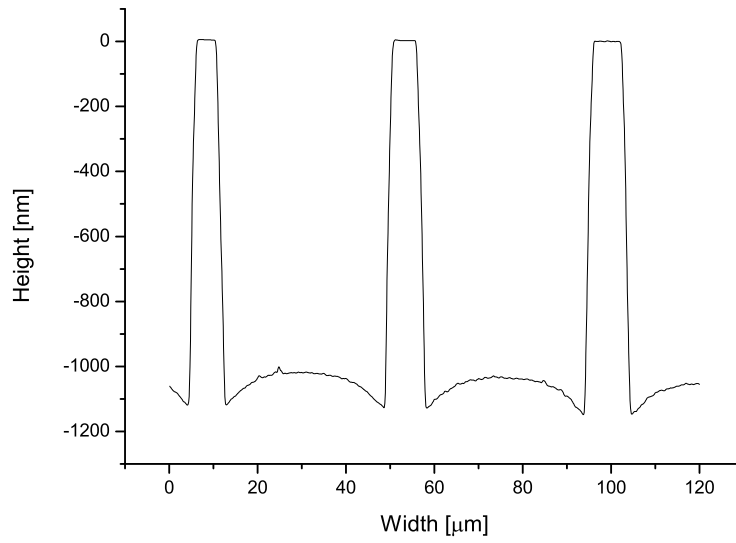
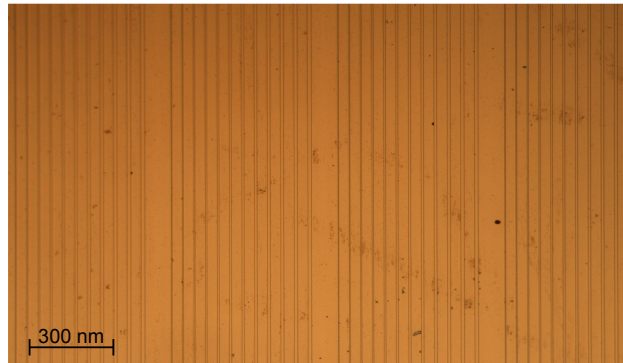


Figure 3.3: Profiler measurement of etched ridges

Figure 3.4: Microscope image of etched ridges in LiNbO_3 from the top

already reducing the etching rate of the crystal by about 50 nm/min.

At the beginning of each RIE process I had to ignite the plasma at higher chamber pressures. The problem was that the ICP RIE had difficulties with keeping the chamber pressure low after about 10 min and the plasma turned off. It reignited again after a few seconds, but was not stable over time. This caused breaks in the RIE process, which were undesirable, yet unavoidable in practice.

The difference of the etching selectivity between CLN and PE CLN and their different

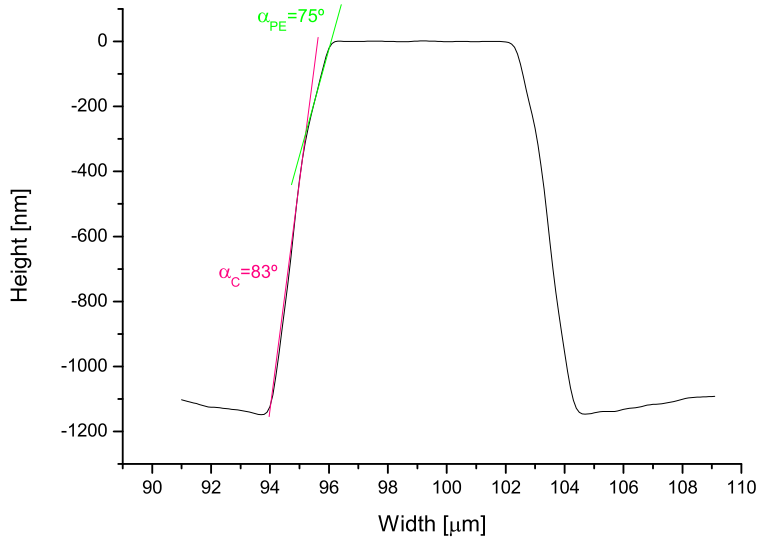


Figure 3.5: Profiler measurement of a ridge with different slope angles in PE CLN (α_{PE}) and CLN (α_C)

faces was rather low. The etching of the -Z face was only very slightly faster and the selectivity was between $s_{PE} = 6$ for PE CLN, $s_C = 5.5$ for CLN and $s_{MgO} = 5$ for PE MgO-doped CLN, respectively. The resulting selectivity for PE CLN is less than what is shown at the beginning in table 3.2, which means there is a time dependency in the process (probably caused by the turn off of the plasma). Another difference occurs in the slope of the etched ridges. Figure 3.5 shows a profiler measurement of a ridge in $LiNbO_3$ with a PE layer of 350 nm. The slope in the PE layer has an angle $\alpha_{PE} = 75^\circ$, which is lower than in CLN, where $\alpha_C = 83^\circ$. Since the slope angle is limiting the smallest structures that can be fabricated this might suggest to avoid the preliminary PE for fabricating nanostructures, although PE improves the etching rate. However, the achieved slope angles are already much better than in other published work, where the slope angles were typically 60° .

The biggest difference was obtained for APE CLN. Etching APE CLN directly gave very poor results both in terms of etch rates and surface roughness. The etched surface turned out to be extremely rough, most likely due to the many different phases present in annealed $Li_{1-x}H_xNbO_3$ [23]. This has been experienced already during wet etching

in former research [29] and is also the main reason why I always chose to perform the annealing of the PE layer after the etching in my subsequent work.

3.2 Study of Annealed Proton Exchanged Layers

As it was described in the methods part about the proton exchange (PE) in section 2.3 a, PE layers should be annealed for obtaining more efficient devices (i.e. lower propagation losses, higher electro-optic and nonlinear coefficients). To find the right conditions, the annealing process has to be first fully characterized. Proton diffusion during annealing is typically nonlinear, which complicates the analysis. Yet this is a critical step for waveguide optimization. The approach I followed in my work was to start from existing nonlinear models for proton diffusion during annealing in CLN, which I used to fit the results of optical experiments made on slab APE waveguides. This allowed to determine the empirical parameters that characterize the properties of the APE waveguides. From this empirical model it was possible to predict the refractive index profile evolution with annealing time, which was then used for subsequent calculations of the guidance properties in the waveguide optimization process. Such a study was developed for APE on both CLN (where similar studies have already been performed) and MgO-doped CLN (where no literature is available).

The experimental basis for the APE modeling was provided by the systematic characterization of annealed PE layers by the prism coupling method, which allows a direct measurement of the mode effective indexes to be used for subsequent reconstruction of the refractive index profile.

Prism Coupling

The characterization has mainly been done with the help of a commercially available prism coupler (Metricon Corporation), which is equipped with a Helium-Neon laser (633 nm). The principle of the prism coupling method is given in figure 3.6. A prism is put into contact with the waveguide that should be analyzed. The refractive index of the prism n_p should be higher than the refractive index of the waveguide. The laser beam is shone under different angles on the prism and then reflected at the prism base due to total

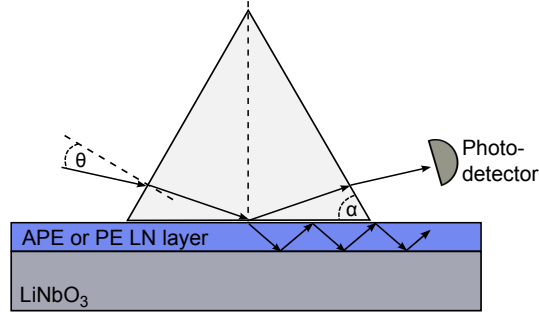


Figure 3.6: Illustration of prism coupling principle

internal reflection, unless there occurs a coupling through the evanescent field in the air gap between the prism and the guiding layer. For the coupling to occur a phase-matching condition needs to be fulfilled, namely that the projection of the wavevector in the prism on the waveguide axis equals the propagation constant of a guided mode ($\beta_{(m)}$). That is why a detector is measuring the intensity of the reflected laser beam. When some power is coupled into the waveguide, there is no total internal reflection anymore and so the detected power drops.

The refractive index of a propagating mode of the waveguide can be calculated with the following equation [10]:

$$N_{eff} = n_P \sin \left\{ \alpha + \arcsin \left(\frac{\theta}{n_P} \right) \right\} \quad (3.3)$$

where α is the base-angle of the prism and θ the angle of the incident laser beam. The refractive index profile $n(z)$ consists of a substrate contribution n_S and a spatially varying index increase associated to the incorporation of protons $\Delta n(z)$ such that $n(z) = n_S + \Delta n(z)$. The obtained index data can then be used to reconstruct $n(z)$ with the help of the *inverse Wentzel-Kramers-Brillouin* (IWKB) method [38], if several (≥ 3) N_{eff} are available. The more effective indexes can be used, the more reliable the index profile can be reconstructed. We shall define the thickness of the PE or APE layer by the $1/e$ depth (d_a) of the index increase $\Delta n(z)$.

3.2.1 Characterization

For the characterization I used CLN and MgO-doped CLN substrates. With a first measurement with the prism coupler I determined their exact bulk index which was

$n_{S(CLN)} = 2.218$ for CLN and $n_{S(MgO)} = 2.191$ for MgO:CLN. The CLN samples were then proton exchanged for 1.5 h at 200°C in BA. Assuming an index difference of $\Delta n_{e(CLN)} = 0.120$ from the bulk to the PE layer, a thickness of $d_{e(CLN)} = 0.687 \mu\text{m}$ was estimated from the prism coupling measurement. It should just be mentioned that the Metricon is using another simple method for getting to d_e , which can be applied to the case of step index planar waveguides (based on analytical solutions to the wave equation available in this case). On the other hand, for graded index profiles, as those of APE waveguides, one has to resort to the IWKB algorithm to retrieve the index profile. The MgO:CLN substrates instead were proton exchanged for 2.5 h at 200 °C and assuming an index difference of $\Delta n_{e(MgO)} = 0.129$ led to an estimated thickness of $d_{e(MgO)} = 0.855 \mu\text{m}$ from the prism coupling measurements.

The samples of the two different substrates were then put into an oven to undergo annealing for different times t_i at a fixed temperature $T_a = 330^\circ\text{C}$. At the beginning the time t_1 was one hour. After the annealing step the prism-coupling measurement was done. The sample was then put into the oven again for $t_i = t_{i-1} + 1 \text{ h}$ and the measurement was done again. This was repeated until the total annealing time was $t_a = \sum_{j=1}^i t_j = 54.5 \text{ h}$. I applied this procedure on two sets of samples in parallel to check the reproducibility.

The effective indexes were then used to reconstruct the profiles of the refractive indexes by the IWKB method and the algorithms of White and Chiang (see [38] for more details) in MATLAB. Those programmes were already existing in the group and are therefore not described here in detail. The results for the Chiang algorithm are shown in figure 3.7a and b.

3.2.2 Modeling

To be able to analyze the data and fit them into a model, the index changes at the surface Δn_a and depths d_a have to be normalized to their initial experimental values. Then given the scaling laws of the APE diffusion [10] one can get “universal” curves for the dependence of $\Delta n'_a$ and d'_a on the annealing time versus t_a regardless of the original PE depth d_e and index increase Δn_e . The scaling requires also to use an effective annealing time normalized by dividing the actual annealing time t_a by the square of the initial PE

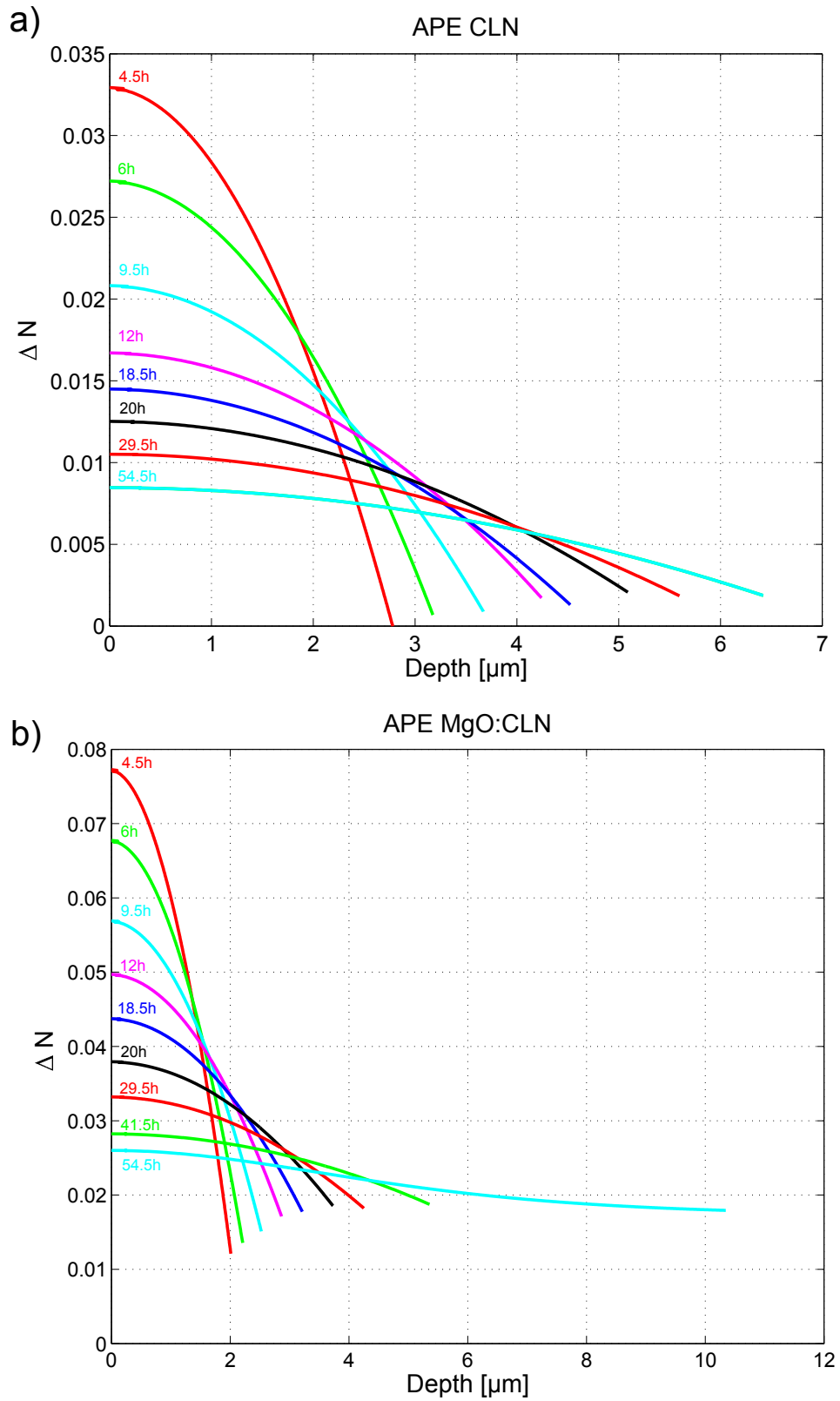


Figure 3.7: Refractive index profiles of CLN (a) and MgO:CLN (b) for different annealing times reconstructed by IWKB method

depth, as follows:

$$\tau = \frac{t_a}{d_e^2} \quad (3.4)$$

where τ is the normalized annealing time. Now figure 3.8 is showing the plots of the normalized surface index changes ($\Delta n'_a = \Delta n_a / \Delta n_e$) as a function of the square root of τ for a) CLN and c) MgO:CLN (blue dots), while $\Delta n_a = \Delta n(0)$. The same is given for the normalized waveguide depth ($d'_a = d_a / d_e$) in figure 3.8b,d. To link the modeling to the experimental data we assume a linear dependence between the proton concentration and the refractive index changes [10]. This relation can be described by:

$$\frac{\Delta n_a(z)}{\Delta n_e(z)} = \alpha \frac{C(z)}{C_0} \quad \alpha \sim 1 \quad (3.5)$$

where C_0 is the proton concentration in the unannealed PE layer and $C(z)$ the proton concentration profile after the annealing. The one-dimensional diffusion equation is then given by [10]:

$$\frac{dC}{dt} = \frac{d}{dz} \left\{ D(C) \frac{dC}{dz} \right\} \quad (3.6)$$

This equation has now to be normalized as discussed before and therefore the expressions $z' = z/d_e$, $C' = C/C_0$ and $\tau = t/d_e^2$ are used to get to:

$$\frac{dC'}{d\tau} = \frac{d}{dz'} \left\{ D(C') \frac{dC'}{dz'} \right\} \quad (3.7)$$

where the normalized proton concentration is $C' = 1$ for the unannealed PE waveguide layer. Assuming a concentration-independent linear diffusion leads to

$$D(C') = D_0 \quad (3.8)$$

The diffusion coefficient D_0 can be calculated from the slope of the regression through the normalized depth data (see green lines in figure 3.8b,d). The calculated results are $D_{0(CL N)} = 0.89 \mu\text{m}^2/\text{hr}$ for CLN and $D_{0(MgO)} = 0.80 \mu\text{m}^2/\text{hr}$ for MgO:CLN.

However, applying this model does not really work very well. A look at figure 3.9 reveals that the linear diffusion model does not fit the experimental data in a proper way. While it might fit reasonably well for 4.5 h the discrepancy between simulation and experiments gets higher for longer annealing times. This is even worse for MgO:CLN. Therefore, the next step was to use a concentration dependent diffusion coefficient (i.e. assume

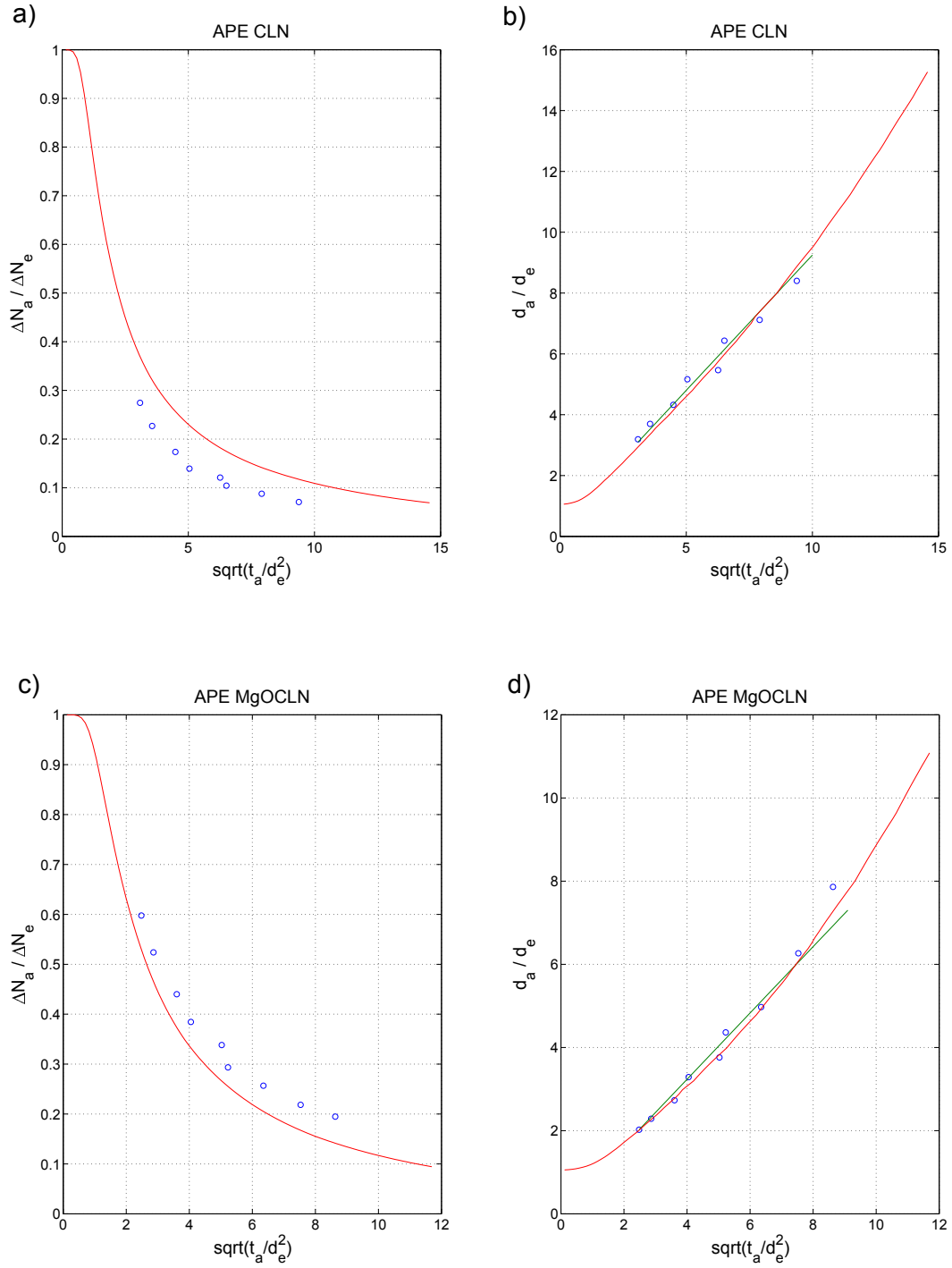


Figure 3.8: Normalized plots of the index increase (a,c) and depth (b,d) including experimental (blue) and fitted (red) data. The green lines were used to calculate the diffusion coefficients from their slopes.

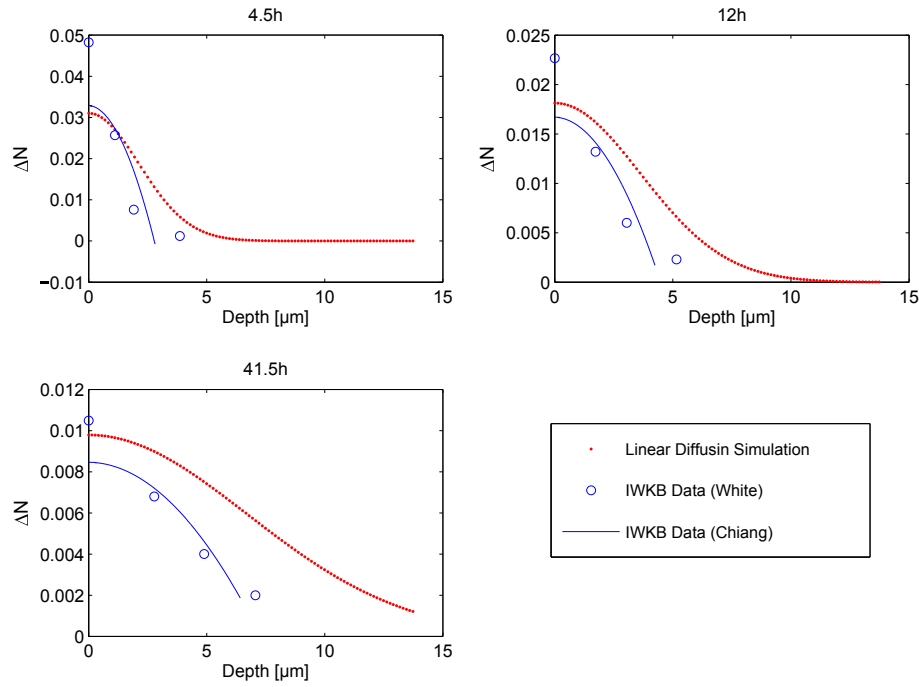


Figure 3.9: Examples of the try test with a linear diffusion model. Comparison between experiments (IWKB index profiles) and simulations (red lines)

a nonlinear proton diffusion during annealing). In light of previous studies I chose a functional dependence of D on C' of the form:

$$D(C') = D_0 \{a \cdot \exp(-b \cdot C') + (1 - a)\} \quad (3.9)$$

where a and b are free parameters to be used to fit the simulations to the experimental data. This means equation 3.7 has to be solved for each t_a with given a and b . Then the simulated profiles and values of $\Delta n'_a$ and d'_a need to be compared to the experimental ones – determined by prism coupling measurements and index profile reconstruction with IWKB. The result of the fit on the $\Delta n'_a$ and d'_a versus τ curves (for all the samples I fabricated and characterized) are shown in figure 3.8. The simulated curves (red lines) are based already upon a fit over all annealing times. In this case the parameters were chosen so to best fit the normalized depth (d'_a) data, instead of the normalized surface index changes, since experimental estimates on the latter contain larger errors (due to the limited number of modes used in the IWKB extrapolation and the IWKB algorithm

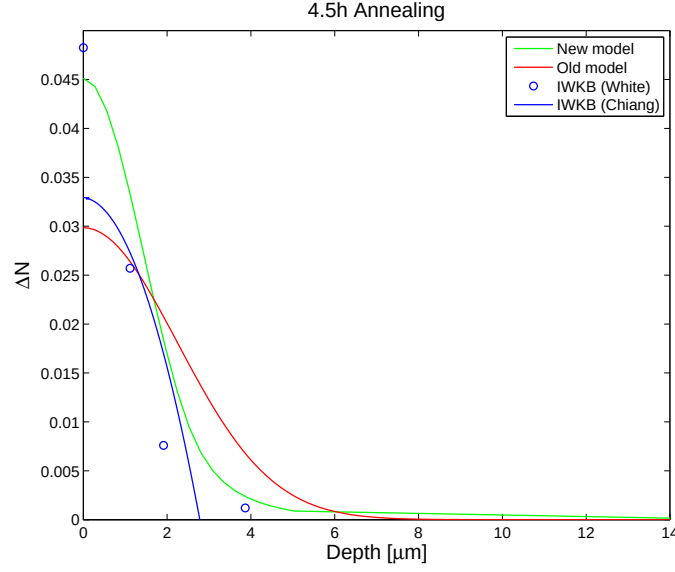


Figure 3.10: Comparison between old and new model with experimental data

itself). The chosen parameters for this simulation are given in table 3.3.

The developed model can now be applied to find the optimal annealing time for a PE

Substrate	a	b
CLN	0.75	40
MgO:CLN	0.81	25

Table 3.3: Parameter for concentration dependent simulations

waveguide with a target thickness d_e . The ultimate aim is to get into the α -phase (see section 2.3), where the value of the surface index change should be around $\Delta n_a = 0.03$ (compare with 2.11).

Unfortunately, the fabrication was already done before the nonlinear model was fully developed and validated. That means the first simulations, on which the optimization was based, were performed with the linear model. However, despite the differences in the index profiles obtained in the linear model and its subsequent refined nonlinear version, the former proved already good enough to provide a first guidance for the optimized fabrication of single-mode waveguides. Mode simulations based upon the two index profiles for the case of the ridges proved also to have little impact on the final guidance properties, which depend most critically on the ridge geometry instead.

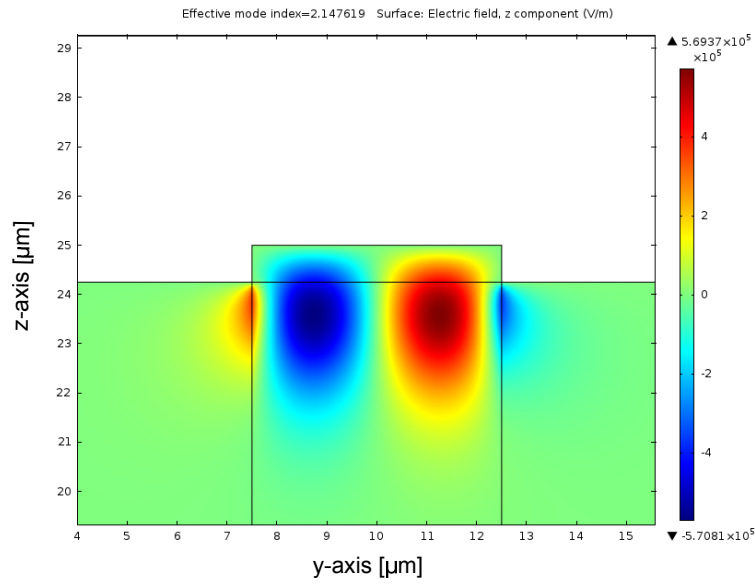


Figure 3.11: Simulation of the mode profile in APE CLN with a PE thickness of 700 nm

3.3 Simulations

Optical measurements on the first fabricated ridge waveguides (PE depth 700 nm) revealed that they were already guiding before the annealing. So, it was clear that one would need to change the PE conditions to achieve single-mode guidance at $\lambda_{FF} = 1550$ nm after annealing. The reason was that annealing leads to higher waveguide depths and as a consequence more guided modes. Simulation confirmed that after annealing those waveguides would have become multi-mode as shown in figure 3.11. Therefore it became obvious that the initial PE depth should be reduced. The question was just how far?

Once we have the index profile of the waveguide we can simulate the guidance properties by solving equation 2.28 in chapter 2. This I did by realizing simulations in COMSOL Multiphysics, where the needed APE profiles have been produced with the fitted (linear) model in MATLAB and were then loaded over a function into COMSOL.

Actually, there is another problem with simulating the index profile according to the fabrication, because the depth of the annealed index profile is much higher than the etched height of the ridge. Due to the fact that the original idea, etching APE, did not work out in practice (see section 3.1.3) I had to choose to do the annealing after the etching of the

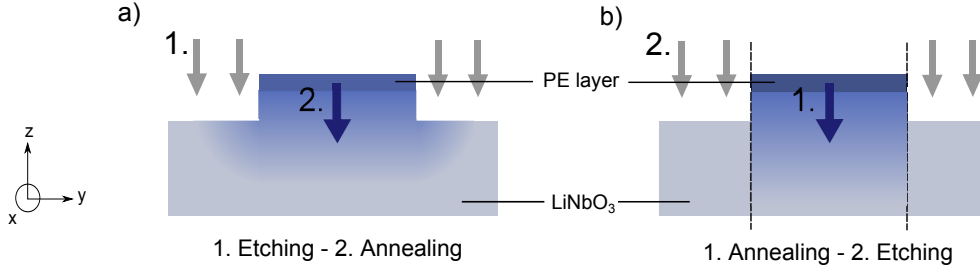


Figure 3.12: Annealing profile according to a) fabrication and b) simplification for simulation

ridge. In this case the annealing profile looks like in figure 3.12a. One can argue that the diffusion model developed in section 3.2.2 would then need to be refined to account for diffusion in a ridge (2D-diffusion model). This is indeed something to be further investigated in future work, but to a first approximation it has a relatively limited impact on the optimizations for single-mode guidance: It is rather unlikely that the designed waveguides switching from single-mode to multi-mode, because the dominant effect is actually associated to the ridge structure, rather than the details of the APE profile. For this reason it is here assumed that the annealing has been done before the etching of the ridge and the used geometry for this simulation looks like in 3.12b instead. The areas at the left and right hand of the dashed lines were replaced with standard $LiNbO_3$ to adjust the model a little better to fit the actual fabrication conditions.

The mathematical model of the refractive index profile in APE CLN was developed at a wavelength of $\lambda_{HeNe} = 633$ nm and as the device we want to fabricate should work at $\lambda_{FF} = 1550$ nm, the generated index profiles need to be corrected according to the dispersion in the APE layer. Also, the substrate index will change, but this can be calculated by the Sellmeier equation (equation 2.41) using the Sellmeier coefficients (table 3.4) given by Jundt/Zelmon [18]. They are giving at the same time the possibility to compute the temperature-dependent dispersion in the crystal, which is very useful, because it is likely for the devices to be heated during the SHG experiments in order to reduce the photorefractive effect. That is why an operating temperature of 80°C was considered here. The dispersion for the APE profiles can be obtained with Sellmeier

Coefficient	n_e
<i>A</i>	2.9804
<i>B</i>	0.02047
<i>C</i>	0.5981
<i>D</i>	0.0666
<i>E</i>	8.9543
<i>F</i>	416.08

Table 3.4: Sellmeier coefficients for the extraordinary index in CLN by Jundt [18]

equations of the form:

$$\Delta n = \sqrt{a_1 + \frac{a_2}{\lambda^2 - a_3^2}} \quad (3.10)$$

with the coefficients given in table 3.5 given by Bortz [10]. The calculation of the sub-

Coefficient	PE CLN	APE CLN
a_1	7.43×10^{-3}	3.43×10^{-5}
a_2	2.64×10^{-3}	1.10×10^{-5}
a_3	0.336	0.326

Table 3.5: Sellmeier coefficients for PE and APE CLN by Bortz [10]

strate index was realized by calling a MATLAB function in COMSOL. The dispersion correction for the profile was already done within the same function that is creating the APE profile.

The ridge had a height of 750 nm in this simulation, because this was the height that was reliably achieved by the etching. The width ranged from 1 to 12 μm . Those were the sizes of the used lithography mask. One can see how the geometry looked in COMSOL in figure 3.13. The computations revealed single-mode guidance with APE profiles whose PE thicknesses (d_e) ranged from 200 to 500 nm and ridge widths from 2 to 8 μm . An example is given in figure 3.14, where the PE thickness was 350 nm and its annealing time 1 h.

Calculations of grating periods for QPM

Furthermore the simulations enabled also the calculation of specific parameters for SHG experiments. The effective indexes for the TM_{00} modes obtained from the simulations - designated as N_{FF} and N_{SH} for the second harmonic, respectively - were used to calcu-

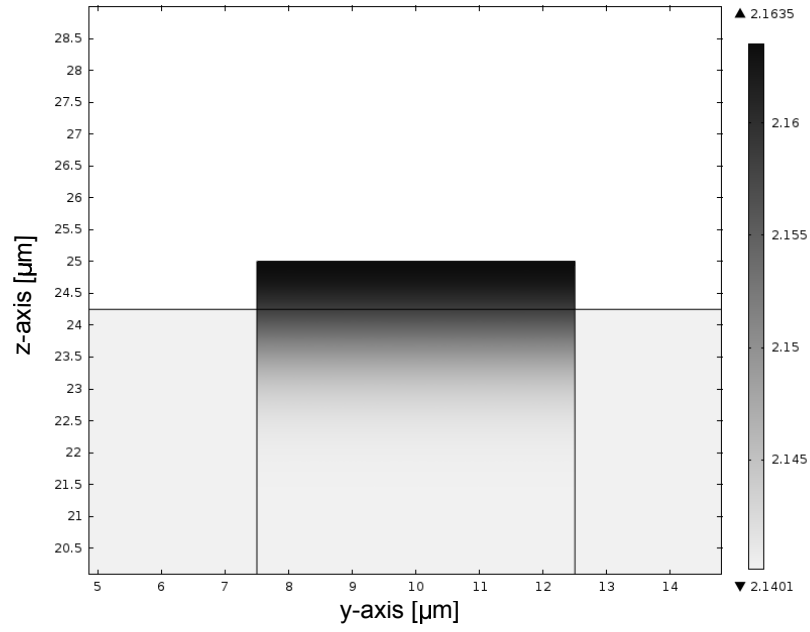


Figure 3.13: Geometry and refractive index profile in the COMSOL simulation (PE thickness of 350 nm, annealing time 1h)

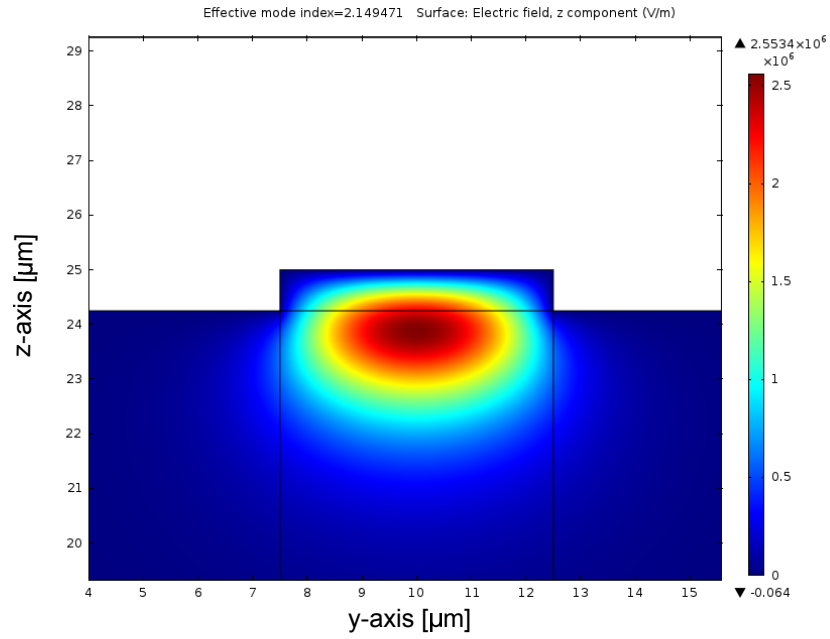


Figure 3.14: Simulation of the mode profile in APE CLN with a PE thickness of 350 nm

late the grating periodicity Λ required for the electric field poling, according to equation 2.36. Due to a lack of time, we could not make ad-hoc masks for the poling and we had to use instead an existing mask with a grating period that was as close as possible to the calculated values. This is one reason why in combination with the mode profiles a width of the ridge of $5\text{ }\mu\text{m}$ was favored. For this dimension the poling period had to be about $\Lambda = 15\text{ }\mu\text{m}$ and this mask was available. The exact values will be given in an overview in section 3.4.

Calculation of the overlap

The calculation of the overlap for SHG required the mode profile to be transferred from COMSOL to MATLAB. The mode profile was therefore written into a file and then loaded by MATLAB to compute the overlap with the help of equation 2.33. The exact values will be again given in the overview of the next chapter.

3.4 The Final Design

All the simulations, calculations and optimizations make it now possible to present details about the most important fabrication parameter listed in table 3.6. However, due to the lack of data on the dispersion of APE MgO:CLN - necessary for the poling grating design - we concentrated only on PPLN ridges. Also the first design with a PE layer thickness of 200 nm was rejected. Its short processing times in terms of PE and annealing would be fairly error-prone. The estimated optimum dimensions for the ridges were $5\text{ }\mu\text{m} \times 750\text{ nm}$.

Substrate	PE layer	PE time	Annealing	Grating PP	Overlap
CLN	200 nm	7.6 min	24 min	$14.8\text{ }\mu\text{m}$	0.5518
CLN	350 nm	23.16 min	1 h	$14.71\text{ }\mu\text{m}$	0.5465
CLN	500 nm	47.27 min	2 h	$14.93\text{ }\mu\text{m}$	0.5308
MgO:CLN	200 nm	12 min	2 h	-	-
MgO:CLN	350 nm	36.75 min	6 h	-	-
MgO:CLN	500 nm	75 min	11 h	-	-

Table 3.6: Final device details for fabrication and SHG experiments

Once again a short explanation of table 3.6: The design is mainly based on the PE layer thickness, which should guarantee single-mode guidance at $\lambda_{FF} = 1550$ nm after annealing. The PE time is the time of proton exchange that is needed to achieve a given thickness of the guiding layer and the annealing time gives the time for realizing an index profile so that one is operating in the α -phase of $Li_{1-x}H_xNbO_3$. In view of the SHG experiments the table is showing also the grating period for the periodic poling required for each design. The overlap is giving an important factor to maximize SHG efficiencies, yet values close to unity are essentially impossible to realize because of dispersion, which yield different confinement properties for the FF and SH.

Chapter 4

Device Fabrication and Characterization

This chapter will now describe the final fabrication of the devices according to the two designs chosen at the end of the last chapter. After all the process steps the samples were inspected by profiler measurements, optical and atomic force microscopy . The results will be shown in the second section. Additionally, the optical setup for the SHG experiments is presented and the first nonlinear optical measurements on the device are described before getting to the discussion of its results at the very end.

4.1 Details of the Fabrication

The devices have been fabricated on the -Z face of congruently $LiNbO_3$ (CLN) and all process steps are summarized and depicted in figure 4.1. The fabrication started with cutting the CLN wafer, which we got from Castech Inc., into small pieces (10 x 16 mm). After a proper cleaning with acetone and isopropanol the samples were prepared for the poling. Therefore the positive photoresist S1818 was spincoated and all samples were put into an oven for baking the resist. The mask which was used for the photolithography had a grating structure with a periodicity of 15 μm . The exposure was done with a Karl Süss mask aligner model MJB3 and afterwards the samples were developed. The electric field poling (EFP) was done by using a gel electrolyte to achieve an electrical contact to

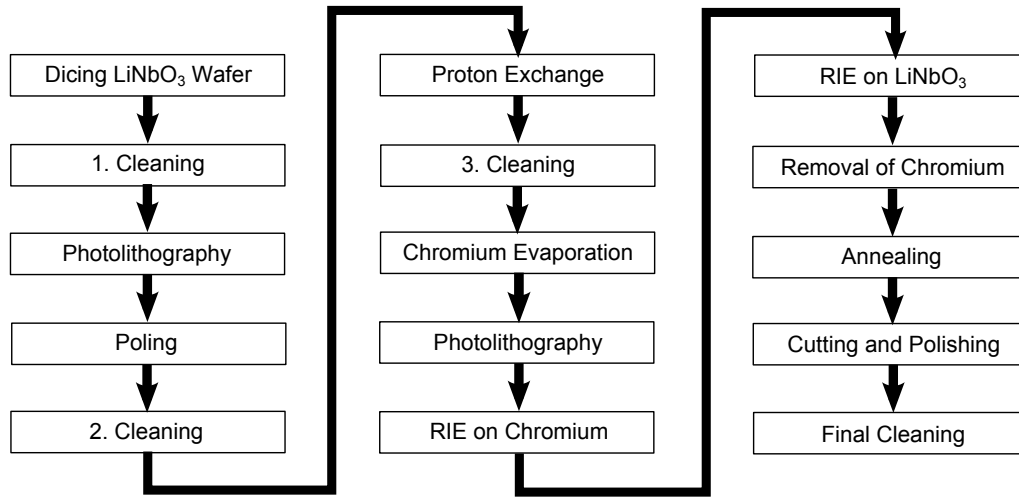


Figure 4.1: Steps of whole device fabrication process

the crystal. Since the coercive field is $E_C = 21$ kV/mm and the thickness of the wafer was $d_W = 0.5$ mm, the voltage for the poling had to be $V_{ext} > E_C \cdot d_W = 10.5$ kV. The sample itself is normally not conductive. To switch an area A , a charge Q (equation 2.44) has to be transferred to the crystal. In order to do so, one uses a set of pulses (adjusting their number and duration) until the desired area is poled. An example of a pulse used for the poling is shown in figure 4.2. By measuring the current during the poling one can draw conclusions on the poled area. A hole set of pulses used to pole one sample can be found in the appendix A.

After the poling the samples were put into the PE setup. The PE was performed at 200°C until a thickness of $d_{e1} = 350$ nm on half of the PPLN samples and $d_{e2} = 500$ nm on the rest of the PPLN substrates was achieved. At that point the ridges were etched according to the etching process in section 3.1. As soon as the ridges were etched into the crystal and the chromium was removed, the PE layer was annealed according to the parameters in table 3.6 at a temperature of 330°C . Now the samples were ready to be cut at both ends of the ridges ($l = 10$ mm) and the ends were also polished to optical quality. As those last processes required to use wax for mounting the samples, a final cleaning procedure was required.

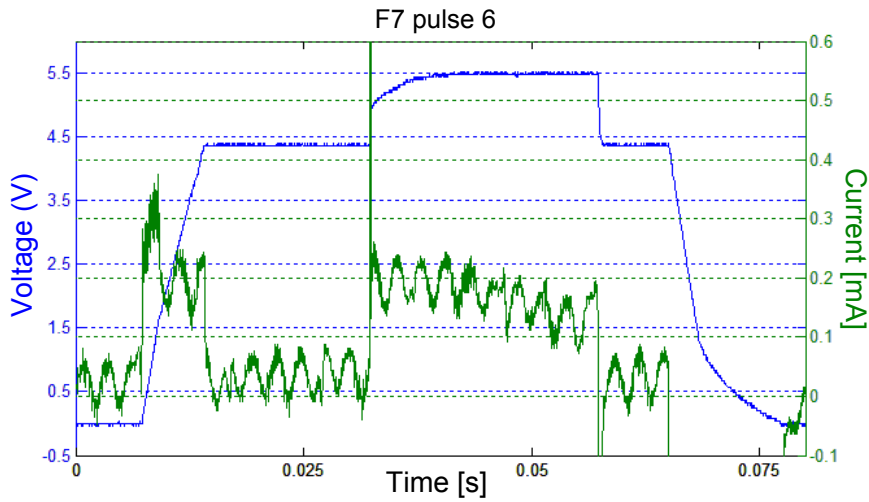


Figure 4.2: Example for poling pulse (blue) and measured current (green) of sample F7. The poling pulse is measured at the signal generator and it is amplified by a factor of 2000.

4.2 Process Inspection

Fabrication always needs a validation of the process and apart from the profiler measurements, which were giving information about the etched structures (described in section 3.1), primarily optical microscopes were used. Figure 4.3 shows a magnification of the fabricated device from the top. The size of the visible ridges ranges from 2 to 12 μm . Unfortunately, it was not possible to develop the 1 μm -ridge from the contact mask correctly during the photolithography, because of the limits of the employed mask aligner (resolution 2 μm). On the other hand, the alignment of ridges and poling grating worked quite well. The next microscope image in figure 4.4 shows the polished end-facets of the device. There were no scratches and the surface had a nice quality.

With the help of Michele Manzo, I made also measurements with an atomic force microscope (AFM), where a very tiny mechanical probe, called cantilever, is moved by piezoelectric scanners over a surface to detect its topography. Using the AFM in contact mode means that the cantilever is brought so close to the surface that it is bended by the force from electrons in the atomic shell of surface atoms and in the case of LiNbO_3 one might also have an electro-static component arising from bound polarization charges. A laser is focused on the cantilever and the reflected beam detected by a photodiode to

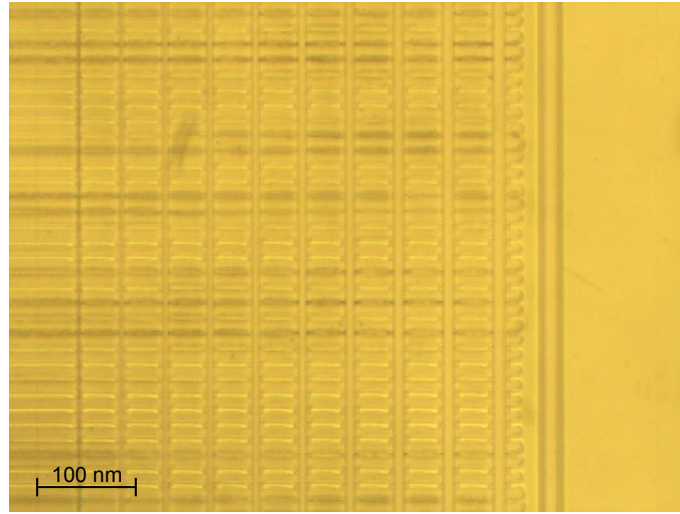
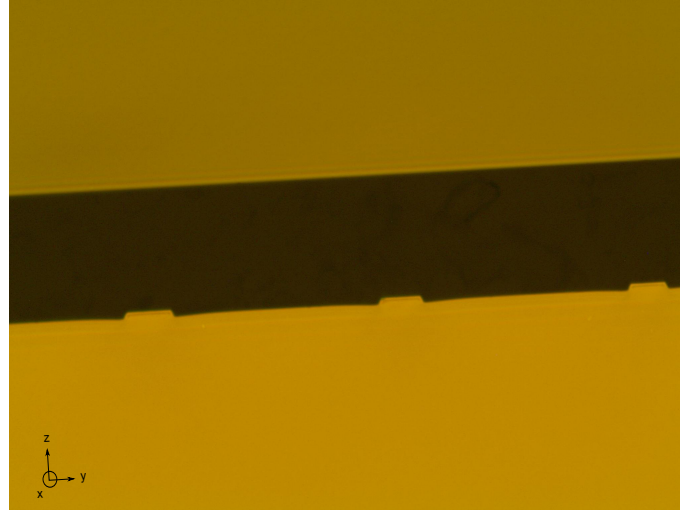


Figure 4.3: PPLN and ridges after etching

Figure 4.4: Microscope image (50x magnification) of etched ridges in LiNbO_3 from the side after cutting and polishing

detect its displacement. With a feedback from the signal from the photodiode, either the height or the force on the cantilever can be kept constant. At the same time one gets images based on the height and deflection of the cantilever. The cantilever we used, was made of silicon nitride, had a resonance frequency of $\omega_0 = 170 \text{ kHz}$ and a spring constant of $k = 45 \text{ N/m}$. This kind of cantilever is actually used for the tapping mode (for contact mode $k = 10 \text{ N/m}$). For our purposes it is better to have a stiff cantilever, because LiNbO_3 is pyroelectric and easily generates charges, which would cause the cantilever

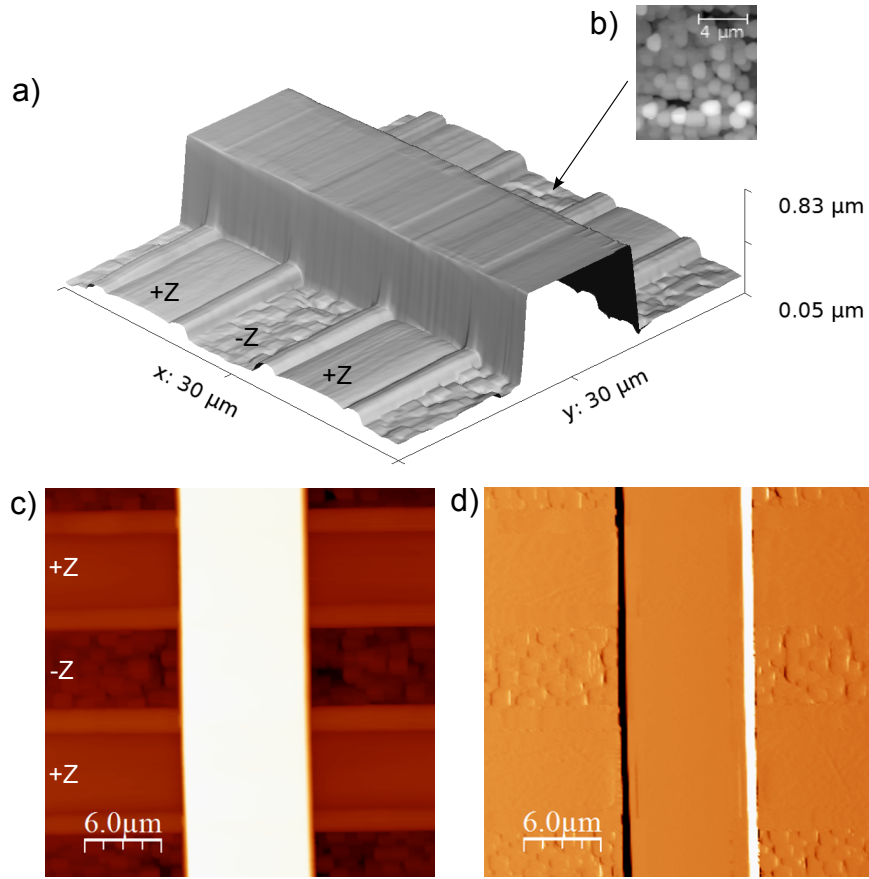


Figure 4.5: AFM measurements of a ridge (8 μm) in CLN with periodically poled domains. a) 3D image, b) detail of the granular features appearing grains on the etched -Z surface (average diameter of a grain 1.4 μm) c) 2D image, d) 2D rendering based on the deflection signal

to stick at the surface. The measurements of an etched PPLN ridge are shown in figure 4.5. As the +Z and -Z face of the crystal are etched differently, you can also see the difference between the poled and unpoled areas resulting in different heights. The image reveals also a perturbation of the etching (little bumps) at the edges of the domain walls. In figure 4.6 you can find instead an AFM image of poled and normal etched domains. There you can see even clearly the boundaries between -Z and +Z domain.

The high resolution AFM measurement from figure 4.5 has been also employed to determine the roughness values of the etched structures, which are given in table 4.1. The most important roughness values (given in root mean square, RMS) for the performance of the device are the ones on top of the ridge and on its sidewalls. A rough surface leads

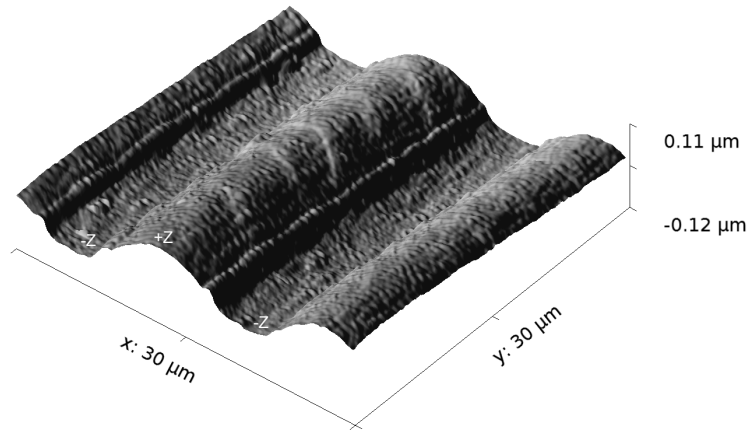


Figure 4.6: AFM measurement of etched PPLN with clearly visible domain walls

Location	RMS
Ridge, top	1.9 nm
Ridge, sidewall	5.0 nm
Domain -Z	14.5 nm
Domain wall +Z	0.8 nm
Unpoled +Z	1.3 nm

Table 4.1: Roughness values

to higher propagation losses in the waveguide. The value measured on top of the ridge corresponds to unetched LiNbO_3 , after removal of the residual chromium mask. The +Z face looks quite smooth after the etching. Only the -Z exhibits a much higher roughness, but since no light is supposed to be guided there, this value is not of high importance.

4.3 Optical Setup

Performing optical measurements with the fabricated devices required a suitable optical setup. Since the waveguides are tiny one has to use objectives, which can be moved with a high precision to focus a beam on the facet of the waveguide in order to couple light into it. One objective is required for coupling light into the device and one for coupling it out again. I started by characterizing the mode profiles of the first unpoled waveguide devices. To this aim I employed a HeNe laser, because it is easier to align an optical setup with a visible laser beam. Its light was coupled into a fiber and

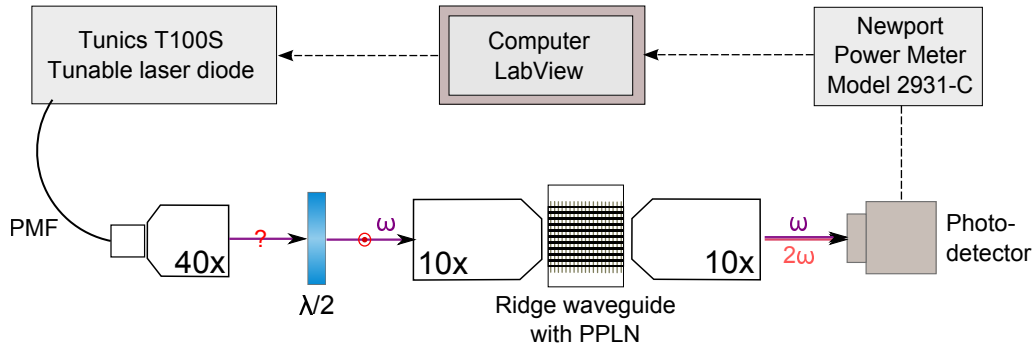


Figure 4.7: Schematic illustration of the experimental setup

then coupled out again in a way so to get a collimated beam, which was coupled into the waveguide by a 10x objective (NA 0.25). Since the devices were processed on Z-cut LiNbO_3 they were only guiding the TM mode and therefore a $\lambda/2$ -plate was used to filter the other polarisation components, before coupling the beam into the waveguide. The outcoupled light (10x objective, NA 0.25) was then analyzed on a screen. This setup was in principle comparable to the bottom part of the schematic illustration in figure 4.7. Thereafter the laser was changed to a tunable diode laser (Yenista Tunics T100S, wavelength range: 1520-1630 nm), which was operating now at $\lambda_T \cong 1550$ nm. Its output was guided by a polarization maintaining fiber (PMF) to the setup and the position of the coupling objectives had to be now further adjusted to optimize the coupling into our devices. Through these first characterizations I found out that the first fabricated PE waveguides were already supporting a guided mode at infrared wavelengths before the PE was annealed. As discussed in chapter 3 this means that the annealing step made the waveguides multi-mode and that the PE depth had to be reduced in subsequent fabrication cycles to optimize the device performance.

After that, the newly designed waveguides with PPLN were characterized. They were annealed according to the improved fabrication conditions outlined in table 3.6. The coupling of the infrared laser beam was more challenging given their higher confinement properties, though still possible.

Then the optical setup was upgraded as depicted in figure 4.7 to allow measurements on the second-harmonic. The outcoupled light from the device was now measured by a photodiode detector (Silicon) linked to a Newport Power Meter (Model 2931-C), which

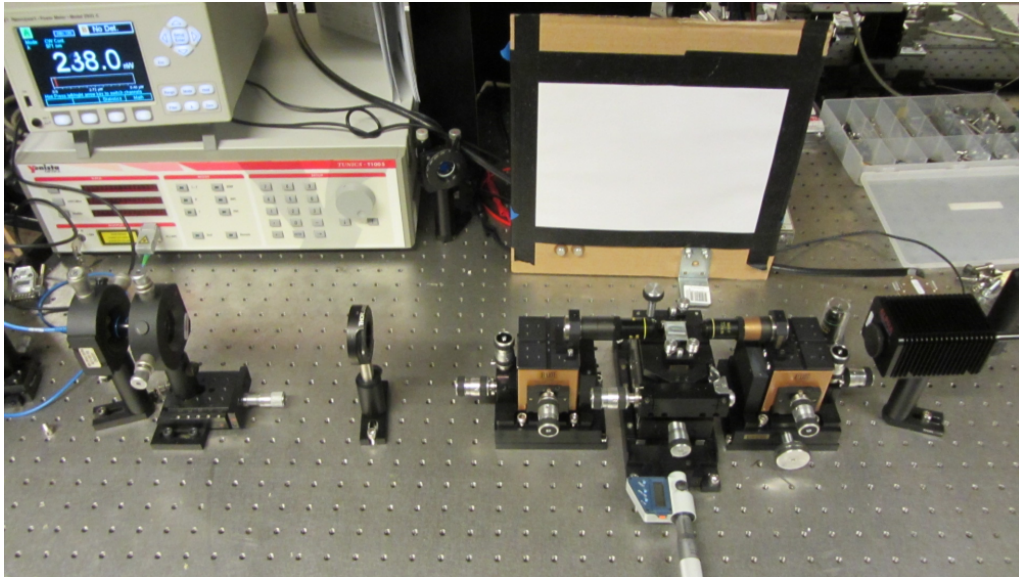


Figure 4.8: Picture of the optical setup for the linear and nonlinear waveguide characterization in the lab. In the back at the left side you can see the power meter and the tunable laser source. In front of it you find the objective that is collimating the beam from the fiber. Next, a $\lambda/2$ -plate and on the right the waveguide station with incoupling and outcoupling objective and an infrared camera at the very end.

detected light up to a wavelength of $1.1\ \mu\text{m}$ and thus automatically filtered off the pump throughput. Both the tunable laser source and the power meter were connected to and controlled by a computer. LabView was used to collect the data from the power meter (measured wavelength = $\lambda_T/2$) and for tuning the wavelength of the laser in its whole range. The results of these measurements are so called SHG tuning curves with the fundamental wavelength on one axis and the detected second-harmonic power on the other one. In such a measurement, a peak at a specific wavelength would show up in case of SHG. Even though the poled grating was actually calculated for one specific wavelength (section 3.3) there are uncertainties in such calculations (specifically concerning waveguide dispersion), which require to search for the correct matching wavelength in a broader spectral range. The wavelength steps were about $\Delta\lambda_T = 0.01\ \text{nm}$, which I have chosen because I expected a theoretical width of the SHG tuning curve of $1\ \text{nm}$ and I wanted to sample it with 100 points.

Despite a few attempts of these measurements with the poled devices, I could not detect any second-harmonic. In fact, I realized that the power of the beam coupled into the

device was too low. While the tunable laser source was delivering 10 mW, I measured only about $P_{FF} = 10 \mu\text{W}$ between coupling objective and my device. If we consider the following equation for calculating the second-harmonic output power P_{SH} :

$$P_{SH} = \eta_{nor} L^2 P_{FF}^2 \quad (4.1)$$

where $\eta_{nor} = \kappa^2$ is the normalized efficiency and L the interaction length, we would only expect to measure 100 pW for P_{SH} (assuming $\eta_{nor} = 100 \text{ \%}/\text{Wcm}^2$ which is close to ideal). This value would have been too low to be measurable with our power meter. Thus, this experimental setup will need some further improvements. If it would not be possible to overcome this problem just by improving the coupling, one could also build an amplifier chain to increase the input power or use a single photon detector for the SHG detection. Figure 4.8 shows a picture of the setup in the lab. In this particular image the photodiode of figure 4.7 is replaced by an infrared camera, which retrieves the mode profile of the outcoupled beam.

4.4 Results and Discussion

An optimal etching recipe for the structuring of LiNbO_3 with RIE was found, which enables the fabrication of ridge structures up to a depth of about 1 μm and with a slope that ranges between 75° (PE CLN) and 83° (CLN). At the same time the etched sidewalls of the ridges possess a surface roughness which almost reaches the quality of optically polished surfaces.

Next, the PE and APE steps were optimized to fabricate waveguides with a highly efficient single-mode guidance at $\lambda = 1550 \text{ nm}$. Therefore the annealing process was modeled and the results were used to generate profiles of the refractive index, which were employed for simulations of the transverse mode profile in the waveguide. These simulations also enabled the calculation of the the poling period and the overlap, which were required to fabricate devices for SHG. Thus, the full design of a ridge waveguide device with PPLN could be achieved and implemented.

The chosen design also led to the full fabrication of the device including all prior optimizations, and an automated optical setup was built up to perform linear and nonlinear

experiments on these devices. It was possible to couple light into the device and to assess its single-modedness, but systematic linear characterizations (mode profiling, determination of coupling losses) are missing due to a lack of time. Instead, first attempts of the nonlinear experiments were performed, which revealed that the optical setup requires further improvements to measure second-harmonic generation from the device. These improvements concern the coupling, the source (boosting the input powers by using an amplification stage) and/or to the detection side (by employing more sensitive detectors). Regarding the fabrication it should be mentioned that the poling was not completely optimized for this application. Better poled structures should be produced in a next round, which can be achieved by using purpose-made grating masks and maybe also metal electrodes rather than just using the photoresist for masking the electric field during the poling.

Producing even better waveguides could be achieved by etching deeper into the crystal, which should be possible in the future. On the other hand, it is desirable also to scale down the waveguide widths to the sub-micron range. Since the observed sidewall angles of the etched ridges are a lot higher than in former work, it should not be a problem to produce these structures into CLN and even MgO:CLN once a higher resolution lithography process is available (e.g. nanoimprint, e-beam, deep UV lithography). With the tools developed in this project it will then be easy to design suitable APE layers and poling gratings.

The first simulations to optimize the process in this direction show that it would be only possible to guide light at $\lambda = 1550$ nm in ridges down to a width of $2\text{ }\mu\text{m}$. This is already a great improvement when comparing with normal channel waveguides where the widths are about $6\text{--}8\text{ }\mu\text{m}$. For nonlinear optics this means an increase by four in the effective areas and according to equation 2.32 therefore also an increase by four in efficiency. To push the technology further, the fabrication process should not be on bulk like here, but on thin film LiNbO_3 , bonded on a lower index material. This would enable to do further scalings of the ridge dimensions to the submicrometric range. The resulting LN nanowires could this time increase the confinement by a further order of magnitude.

A last comment is devoted to an evaluation of the suitability of the developed process for the fabrication of Bragg gratings and photonic crystal structures for more advanced

LiNbO₃ waveguides. Bragg gratings for example would require periodicities of about 400 nm. With the current slope angles resulting after the etching, it would be possible to produce structures with limited heights of 373 nm (for $\alpha = 75^\circ$ - PE CLN) and 814 nm (for $\alpha = 83^\circ$ - CLN) until the sidewalls of the ridges are merging at the bottom. This gives also an indication of the importance of the slope angle for the structuring when scaling to submicron scales.

Chapter 5

Conclusions and Future Work

A ridge waveguide-based nonlinear optical device in congruent $LiNbO_3$ (CLN) has been fabricated. The substrate material has been periodically poled and then ridges were etched by an optimized RIE process using a chromium hard mask. Annealed proton exchange (APE) was employed in order to achieve vertical confinement. The results from the optimized RIE recipe, yielding quite smooth and steep etched slopes (ridge wall roughness ~ 5 nm and angles up to 83°), suggest that it would be possible to etch even structures with widths in the sub-micron range, suitable for the demonstration of photonic nanowires in $LiNbO_3$.

Furthermore, during this study an empirical model was also developed for the annealing of PE layers in both CLN and MgO:CLN. The model describes the redistribution of protons in the substrate during annealing as a nonlinear diffusion process and made it possible to predict the evolution of the APE refractive index profiles as a function of the process conditions. It therefore provided an extremely valuable tool for the optimization of the PE and annealing conditions. The goal of this optimization was to achieve high-confinement single-mode waveguides at telecom wavelengths ($\lambda \cong 1550$ nm) suitable for efficient SHG. Accordingly, the fabrication process was coupled with numerical simulations to evaluate the evolution of the APE index profiles as a function of fabrication conditions and then perform modal analyzes to estimate the guidance properties of the corresponding APE ridge waveguides.

It was also shown that the chosen APE ridge waveguide enables shrinking the waveguide

lateral dimension down to about $2\text{ }\mu\text{m}$, while still guiding light around $\lambda = 1550\text{ nm}$. A comparison with current state-of-the-art channel waveguides, having typical widths of $6\text{--}8\text{ }\mu\text{m}$ indicates a potential improvement by a factor of four of SHG efficiencies.

Therefore the last part of the work concentrated on combining the optimized APE ridge devices with the periodic poling and building a first optical setup for their nonlinear characterization. Due to the time constraints of this thesis project we managed to perform only a first attempt of testing SHG in the optimized devices, which was not successful due to the limited excitation powers available experimentally. Work is currently under way to upgrade the optical setup to enable 100 mW pumping of the APE PPLN ridges, from which we expect an output power of 10 mW at the second-harmonic.

During the period of this project, a similar PPLN ridge device (for SHG from $1.064\text{ }\mu\text{m}$) was demonstrated by Sun and Xu [31]. Nevertheless it is worth underlining that their ridges were still relatively wide ($8.5\text{ }\mu\text{m}$), due to the limitations of the process used for their fabrication - namely cutting with a diamond saw. On the other hand, such limitations do not apply to the dry etching method used here, which allows instead the fabrication of much finer structures, yielding even a higher confinement. Moreover, by bonding a thin film layer of LiNbO_3 on a lower index material, one can expect to be able to guide light at telecom wavelengths even in sub-micron ridges. After changing the lithography techniques (e.g. to e-beam), the developed RIE recipe can be employed straight away to fabricate such waveguides.

In a further fabrication cycle the substrate material should also be changed to MgO:CLN .

Bibliography

- [1] R. S. Weis and T. K. Gaylord. Lithium Niobate: Summary of Physical Properties and Crystal Structure. *Applied Physics A*, 37:191–203, 1985. [cited at p. 1, 13, 14, 17, 67]
- [2] A. Räuber. *Chemistry and Physics of Lithium Niobate*, volume 1, chapter 7. North-Holland Publishing Company, 1978. [cited at p. 1, 13, 16]
- [3] M.-C. Oh, Hua Zhang, C. Zhang, H. Erlig, Y. Chang, B. Tsap, D. Chang, A. Szep, W. H. Steier, H. R. Fetterman, and L. R. Dalton. Recent Advances in Electrooptic Polymer Modulators Incorporating Highly Nonlinear Chromophore. *IEEE Journal on Selected Topics in Quantum Electronics*, 7(5):826–835, 2001. [cited at p. 1]
- [4] J. Leuthold, C. Koos and W. Freude. Nonlinear silicon photonics. *Nature Photonics*, 4:536–544, 2010. [cited at p. 1]
- [5] K. Gallo. All-optical signal processing in periodically poled LiNbO₃. In *Optics & Photonics Days, Sweden*, 2012. [cited at p. 1]
- [6] D. Wiersma. Controlling photons with light. *Nature Photonics*, 2:1–2, 2008. [cited at p. 1]
- [7] K. Gallo, G. Assanto, K. R. Parameswaran, and M. M. Fejer. All-optical diode in a periodically poled lithium niobate waveguide. *Applied Physics Letters*, 79:314–317, 2001. [cited at p. 2]
- [8] P. Franken, A. Hill, C. Peters, and G. Weinreich. Generation of optical harmonics. *Physical Review Letters*, 7(4):118–119, 1961. [cited at p. 3]
- [9] R. W. Boyd. *Nonlinear Optics*. Academic Press, 2003. [cited at p. 4, 7]
- [10] M. L. Bortz. *Quasi-Phasematched Optical Frequency Conversion in Lithium Niobate Waveguides*. PhD thesis, Stanford University, 1994. [cited at p. 4, 11, 18, 32, 33, 35, 41, 69]

- [11] F. Laurell. Periodically poled materials for miniature light sources. *Optical Materials*, 11:235–244, 1999. [cited at p. 9]
- [12] J. L. Jackel, C. E. Rice, and J. J. Veselka. Proton exchange for high-index waveguides in LiNbO_3 . *Applied Physics Letters*, 41:607–608, 1982. [cited at p. 10, 17]
- [13] R.V. Schmidt and I.P. Kaminow. Metal-diffused optical waveguides in LiNbO_3 . *Applied Physics Letters*, 25:458–460, 1974. [cited at p. 10]
- [14] G. Destefanis, J. Gailliard, E. Ligeon, S. Valette, B. Farmery, P. Townsend, and A. Perez. The formation of waveguides and modulators in LiNbO_3 by ion implantation. *Journal of Applied Physics*, 50:7898–7905, 1979. [cited at p. 10]
- [15] S. Mailis, C. Riziotis, I. Wellington, P. Smith, C. Gawith, and R. Eason. Direct ultraviolet writing of channel waveguides in congruent lithium niobate single crystals. *Optics Letters*, 28(16):1433–1435, 2003. [cited at p. 10]
- [16] H. Nishihara, M. Haruna and T. Suhara. *Optical Integrated Circuits*. McGraw-Hill, 1989. [cited at p. 11]
- [17] Yu. N. Korkishko and V.A. Fedorov. *Ion Exchange in Single Crystals for Integrated Optics and Optoelectronics*. Cambridge International Science Publishing, 1999. [cited at p. 13, 16, 17, 67]
- [18] D. Zelmon, D. Small, and D. Jundt. Infrared corrected Sellmeier coefficients for congruently grown lithium niobate and 5 mol. % magnesium oxide doped lithium niobate. *Journal of the Optical Society of America B*, 14(12):3319–3322, 1997. [cited at p. 15, 16, 40, 41, 67, 69]
- [19] M. Lawrence. Lithium Niobate Integrated Optics. *Reports on Progress in Physics*, 56:363–429, 1993. [cited at p. 16, 67]
- [20] C. E. Rice. The structure and properties of $\text{Li}_{1-x}\text{H}_x\text{NbO}_3$. *Journal of Solid State Chemistry*, 64:188–199, 1986. [cited at p. 17]
- [21] Yu. N. Korkisho, V. Fedorov, S. Kostritskii, E. Maslennikov, M. Frolova, and A. Alkaev. Proton-exchanged waveguides in MgO-doped LiNbO_3 : Optical and structural properties. *Journal of Applied Physics*, 94(2):1163–1170, 2003. [cited at p. 18]
- [22] M. De Micheli, J. Botineau, S. Neveu, P. Sibillot, and D. Ostrowsky. Independent control of index and profiles in proton-exchanged lithium niobate guides. *Optics Letters*, 8(2):114–115, 1983. [cited at p. 18]

- [23] Yu. N. Korkishko and V. Fedorov. Structural Phase Diagram of $H_xLi_{1-x}NbO_3$ Waveguides: The Correlation Between Optical and Structural Properties. *IEEE Journal of Selected Topics in Quantum Electronics*, 2(2):187–196, 1996. [cited at p. 18, 19, 30, 67]
- [24] M. Yamada, N. Nada, M. Saitoh, and K. Watanabe. First-order quasi-phase matched $LiNbO_3$ waveguide periodically poled by applying an external field for efficient blue second-harmonic generation. *Applied Physics Letters*, 62(5):435–436, 1993. [cited at p. 20]
- [25] L. Myers, R. Eckardt, M. Fejer, R. Byer, W. Bosenberg, and J. Pierce. Quasi-phase-matched optical parametric oscillators in bulk periodically poled $LiNbO_3$. *Journal of the Optical Society of America B*, 12(11):2102–2116, 1995. [cited at p. 20]
- [26] M. Manzo. Influence of Selective Proton Exchange on Periodically Poled Lithium Niobate. Master’s thesis, KTH - Royal Institute of Technology, 2010. [cited at p. 20, 21]
- [27] D. Xue and K. Kitamura. Origin of Differential Etching Rates of the +Z and - Z Faces of Lithium Niobate Crystal. *Ferroelectrics Letters*, 29:89–93, 2002. [cited at p. 21]
- [28] G. Miller. *Periodically Poled Lithium Niobate: Modeling, Fabrication and Nonlinear-optical Performance*. PhD thesis, Stanford University, 1998. [cited at p. 21]
- [29] F. Laurell, J. Webjörn, G. Arvidsson, and J. Holmberg. Wet Etching of Proton-exchanged Lithium Niobate - A Novel Processing Technique. *Journal of Lightwave Technology*, 10(11):1606–1609, 1992. [cited at p. 25, 31]
- [30] T. Kawaguchi, T. Yoshino, J. Kondo, A. Kondo, S. Yamaguchi, K. Noda, T. Nehagi, M. Imaeda, K. Mizuuchi, Y. Kitaoka, T. Sugita, and K. Yamamoto. High-power blue/violet QPM-SHG laser using a new ridge type waveguide. In *CLEO*, 2001. [cited at p. 25]
- [31] J. Sun and C. Xu. 466 mW green light generation using annealed proton-exchanged periodically poled $MgO:LiNbO_3$ ridge waveguides. *Optics Letters*, 37(11):1606–1609, 2012. [cited at p. 25, 58]
- [32] H. Hu, A. Milenin, R. Wehrspohn, H. Hermann, and W. Sohler. Plasma etching of proton-exchanged lithium niobate. *Journal of Vacuum Science Technology A*, 24(4):1012–1015, 2006. [cited at p. 25]
- [33] G. Ulliac, B. Guichardaz, J. Rauch, S. Queste, S. Benchabane, and N. Courjal. Ultra-smooth $LiNbO_3$ micro and nano structures for photonic applications. *Microelectronic Engineering*, 88:2417–2419, 2011. [cited at p. 25]

- [34] D. Jun, J. Wei, C. Png, S. Guangyuan, J. Son, H. Yang, and A. Danner. Deep anisotropic etching with SF_6/Ar inductively coupled plasmas. *Journal of Vacuum Science Technology B*, 30(1), 2012. [cited at p. 25, 27, 28]
- [35] H. Hu, R. Ricken and W. Sohler. Lithium niobate photonics wires. *Optics Express*, 17(26):24261–24268, 2009. [cited at p. 25]
- [36] F. Lacour, N. Courjal, M. Bernal, A. Sabac, C. Bainier, and M. Spajer. Nanostructuring lithium niobate substrates by focused ion beam milling. *Optical Materials*, 27:1421–1425, 2005. [cited at p. 25]
- [37] F. Schrempel, Th. Gischkat, H. Hartung, E. Kely, and W. Wesch. Ion beam enhanced etching of LiNbO_3 . *Nuclear Instruments and Methods in Physics Research B*, 250:164–168, 2006. [cited at p. 25]
- [38] P. Hertel and H. P. Menzler. Improved Inverse WKB Procedure to Reconstruct Refractive Index Profiles of Dielectric Planar Waveguides. *Applied Physics B*, 44:75–80, 1987. [cited at p. 32, 33]

Appendices

Appendix A

Appendix Poling

The following diagrams (figure A.1, next page) are showing all used poling pulses for sample F7. This sample would actually require to pole an area of $A = 0.25 \text{ cm}^2$ and a charge of $Q_P = 2 \cdot A \cdot P_S = 39 \text{ }\mu\text{C}$. By integrating the measured current during the poling MATLAB calculated $Q_P = 3.2415 \text{ }\mu\text{C}$ and $A = 2.0779 \text{ mm}^2$ instead. That means this sample is not completely poled. It was difficult to pole it because it was already proton exchanged.

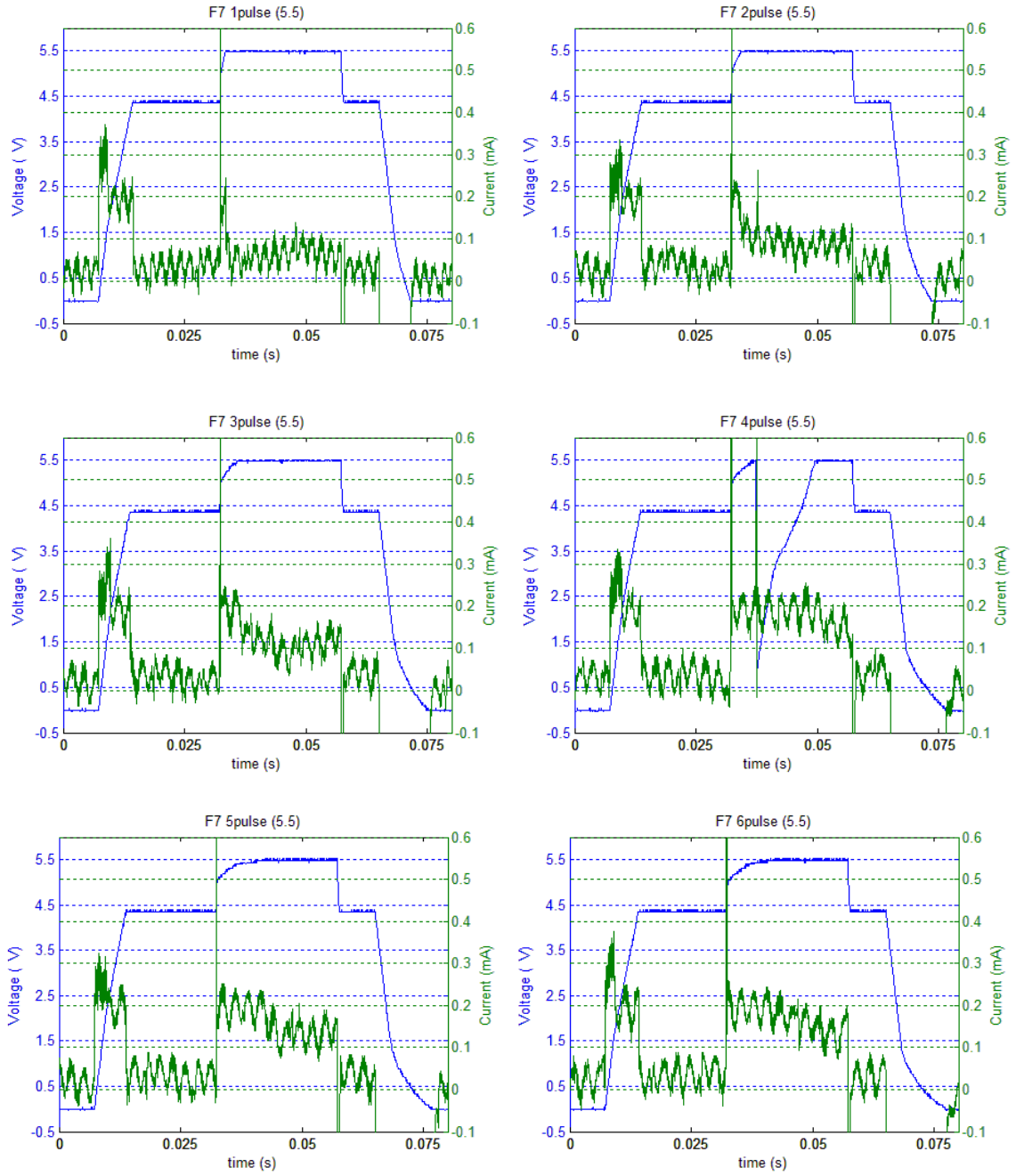


Figure A.1: All poling pulses (blue) and measured current (green) for sample F7

List of Figures

2.1	Sum-frequency generation	6
2.2	a) Geometry of second harmonic generation and b) energy level diagram . .	7
2.3	Periodic engineered polarization for first order QPM	9
2.4	Second-harmonic power growth in a nonlinear medium	9
2.5	a) Buried and b) ridge waveguide structure	11
2.6	Composition phase diagram of $\text{LiO}_2\text{-NbO}_5$ compound [17]	13
2.7	Standard orientation of X,Y,Z axes, after Weis and Gaylord [1]	14
2.8	Dispersion curves for the ordinary and extraordinary index of CLN [18] . .	15
2.9	Transmission spectrum of CLN [19]	16
2.10	Refractive profiles for PE and APE waveguides	18
2.11	Structural phase diagram of $\text{Li}_{1-x}\text{H}_x\text{NbO}_3$ on Z-cut[23]	19
2.12	Schematic of the used PE setup	19
2.13	Schematic of PPLN with stabilized domains	21
3.1	Fabrication of ridges into LiNbO_3	26
3.2	Influence of the temperature on the etching of -Z CLN	28
3.3	Profilometer measurement of etched ridges	29
3.4	Microscope image of etched ridges in LiNbO_3 from the top	29
3.5	Profilometer measurement of a ridge with different slope angles	30
3.6	Illustration of prism coupling principle	32
3.7	Refractive index profiles reconstructed by IWKB method	34
3.8	Normalized plots for experimental and fitted data on the index profiles . . .	36

3.9	Examples of the try test with a linear diffusion model	37
3.10	Comparison between old and new model with experimental data	38
3.11	Simulation of the mode profile in APE CLN (PE thickness 700 nm)	39
3.12	Annealing profile according with simplification for simulation	40
3.13	Geometry and refractive index profile in simulation	42
3.14	Simulation of the mode profile in APE CLN (PE thickness 350 nm)	42
4.1	Steps of whole device fabrication process	46
4.2	Example for poling pulse and current	47
4.3	PPLN and ridges after etching	48
4.4	Microscope image of etched ridges in $LiNbO_3$	48
4.5	AFM measurements of a poled ridge in CLN	49
4.6	AFM measurement of etched PPLN	50
4.7	Schematic illustration of the experimental setup	51
4.8	Picture of the optical setup in the lab	52
A.1	Example of poling	66

List of Tables

3.1	Etching etching rate [nm/sec] with different gases	27
3.2	Etching selectivity with different gases	27
3.3	Parameter for concentration dependent simulations	38
3.4	Sellmeier coefficients for the extraordinary index in CLN by Jundt [18] . .	41
3.5	Sellmeier coefficients for PE and APE CLN by Bortz [10]	41
3.6	Final device details for fabrication and SHG experiments	43
4.1	Roughness values	50